



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

PROJECTS

The domain of the Project:
PHYSICAL DESIGN (VLSI)

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BATCH: G4-25-VLSI



ASSIGNMENT 1: Documentation to capture issues, daily updates and Tool Installation

A. Linux Commands Practice

1. Basic commands: Used to perform file system navigation and file manipulation.

Command	Description
pwd	Print current directory
ls	List files
cd	Change directory
mkdir	Create directory
rm	Remove files
cp	Copy files
mv	Move/rename files

2. VI Editor: used to edit configuration files and scripts used in VLSI flow.

Operation	Command
Open file	vi filename
Insert mode	i
Save	:w
Exit	:q
Save & Exit	:wq



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Operation	Command
Force quit	:q!

3. chmod: Used to control file execution permissions.

Syntax: chmod <permissions> filename

Examples

- chmod +x script.sh
- chmod 755 run.sh

Number	Permission
7	Read + Write + Execute
5	Read + Execute
4	Read only

4. grep: Used to search specific patterns inside files.

Syntax: grep "pattern" filename

Examples

- grep "error" log.txt
- grep -i "warning" log.txt
- grep -r "slack"

5. awk: To process and analyze structured text/log files.

Syntax: awk '{print \$column_number}' file



B. Openlane Environment Learning: Directory structure explore

```
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane$ cd designs
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs$ ls
cl      spm
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs$ cd spm
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm$ ls
config.json pin_order.cfg runs src
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm$ cd runs
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs$ ls
RUN_2026.05.06.12.50.51 openlane_test
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs$ cd openlane_test
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ ls
OPENLANE_COMMIT PDK_SOURCES cmds.log config.in.tcl logs openlane.log reports results runtime.yanl tmp warnings.log
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ cd reports
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ ls
cts floorplan manufacturability.rpt metrics.csv placement routing signoff synthesis
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd placement
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/placement$ ls
11-dpl_sta.checks.rpt 11-dpl_sta.min.rpt 11-dpl_sta.skew.rpt 8-gpl_sta.checks.rpt 8-gpl_sta.min.rpt 8-gpl_sta.skew.rpt
11-dpl_sta.max.rpt 11-dpl_sta.power.rpt 11-dpl_sta.summary.rpt 8-gpl_sta.max.rpt 8-gpl_sta.power.rpt 8-gpl_sta.summary.rpt
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/placement$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd routing
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/routing$ ls
16-rsz_design_sta.checks.rpt 16-rsz_design_sta.power.rpt 18-rsz_tuning_sta.checks.rpt 18-rsz_tuning_sta.power.rpt 21-grt_sta.checks.rpt 21-grt_sta.power.rpt 24-wire_lengths.csv
16-rsz_design_sta.max.rpt 16-rsz_design_sta.skew.rpt 18-rsz_tuning_sta.max.rpt 18-rsz_tuning_sta.skew.rpt 21-grt_sta.max.rpt 21-grt_sta.skew.rpt drt.drc
16-rsz_design_sta.min.rpt 16-rsz_design_sta.summary.rpt 18-rsz_tuning_sta.min.rpt 18-rsz_tuning_sta.summary.rpt 21-grt_sta.min.rpt 21-grt_sta.summary.rpt drt.klayout.xml
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/routing$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd signoff
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/signoff$ ls
26-mca 31-rx_sta.checks.rpt 31-rx_sta.power.rpt 32-irdrop-VDD.rpt 35-xor.xml 41-antenna_violators_pins.txt drc.rpt spice.feedback.txt spm.rpt.error.gz
31-rx_sta.max.rpt 31-rx_sta.skew.rpt 32-irdrop-VDD.rpt 39-spm.lvs.rpt drc.klayout.xml drc.tcl spm.rpt drc.tr spm.rpt.debug.gz
30-mca 31-rx_sta.min.rpt 31-rx_sta.summary.rpt 35-xor.rpt 41-antenna_violators.rpt drc.rdb
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/signoff$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd cts
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/cts$ ls
13-cts_sta.checks.rpt 13-cts_sta.max.rpt 13-cts_sta.min.rpt 13-cts_sta.power.rpt 13-cts_sta.skew.rpt 13-cts_sta.summary.rpt
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/cts$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd floorplan
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/floorplan$ ls
3-initial_fp_core_area.rpt 3-initial_fp_die_area.rpt
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports/floorplan$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/reports$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ ls
OPENLANE_COMMIT PDK_SOURCES cmds.log config.in.tcl logs openlane.log reports results runtime.yanl tmp warnings.log
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ cd results
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ ls
cts Final floorplan placement routing signoff synthesis
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ cd cts
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/cts$ ls
spm.def spm.sdc
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/cts$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ cd final
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/final$ ls
lef gds lef lib mag mag.lef sdc sdf spef spi verilog
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/final$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ cd routing
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/routing$ ls
mca spm.def spm.nl.v spm.oob spm.pnl.v
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/routing$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ cd signoff
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/signoff$ ls
spm.drc mag spm.gds spm.klayout.gds spm.lef spm.lef.mag spm.lef.spice spm.lib spm.lyp spm.mag spm.magic.gds spm.sdf spm.spice
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results/signoff$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/results$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ ls
OPENLANE_COMMIT PDK_SOURCES cmds.log config.in.tcl config.in.tcl logs openlane.log reports results runtime.yanl tmp warnings.log
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test$ cd tmp
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp$ ls
15-spm.def 15-spm.oob 15-spm.sdc 17-spm.nl.v 17-spm.pnl.v cts floorplan merged.max.lef merged.non.lef routing synthesis
15-spm.nl.v 15-spm.pnl.v 17-spm.def 17-spm.oob 17-spm.sdc dimensions.txt layers.list merged.min.lef placement signoff
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp$ cd routing
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp/routing$ ls
19-global.def 19-global.oob 22-fill.nl.v 22-fill.pnl.v global.nl.v groute-congestion.rpt resizer-routing-tuning-congestion.rpt
19-global.guide 22-fill.def 22-fill.oob config.tracks global.pnl.v resizer-routing-design-congestion.rpt
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp/routing$ cd ..
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp$ cd signoff
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp/signoff$ ls
36-spm.nl.v 36-spm.p.def 36-spm.pnl.v 39-setup.file.lef lvs gds_ptrs.mag magic.spice_ext spm.cdl spm.power
lucifer@PhysicalDesign:~/Desktop/OpenLane/OpenLane/designs/spm/runs/openlane_test/tmp/signoff$
```

Fig1.1. Exploring the Directory using cd and ls commands



C. TCL Task

1. Wire Length Calculation

- Find out total Wire length, Clock wire length and Data wire length of design

Method 1- Using Basic commands from 24-wire_lengths.csv file

```
net64,0.46
net76,0.46
lucifero@PhysicalDesign:~/Desktop/openlane/openlane/designs/spm/runs/openlane_test/reports/routing$ awk -F',' '{sum += $2} END {print sum}' 24-wire_lengths.csv
6716.82
lucifero@PhysicalDesign:~/Desktop/openlane/openlane/designs/spm/runs/openlane_test/reports/routing$ ^[[200-grep -i clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'-
awk: line 1: syntax error at or near ~
grep: command not found
lucifero@PhysicalDesign:~/Desktop/openlane/openlane/designs/spm/runs/openlane_test/reports/routing$ grep -i clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
784.48
lucifero@PhysicalDesign:~/Desktop/openlane/openlane/designs/spm/runs/openlane_test/reports/routing$ grep -iv clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
5932.34
```

1. Total wire length: 6716.82

```
awk -F',' '{sum += $2} END {print sum}' 24-wire_lengths.csv
```

2. Clock wire length: 784.48

```
grep -i clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
```

3. Data wire length: 5932.34

```
grep -iv clk 24-wire_lengths.csv | awk -F',' '{sum += $2} END {print sum}'
```

Method 2- Using openroad and Manual calculations

```
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/14-resizer_design.log)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/16-resizer_timing.log)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running Global Routing (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/18-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 19]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/18-global_write_netlist.log)...
[STEP 20]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/20-grt_sta.log)...
[STEP 21]
[INFO]: Running Fill Insertion (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/21-fill.log)...
[STEP 22]
[INFO]: Running Detailed Routing (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/22-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 23]
[INFO]: Checking Wire Lengths (log: designs/spm/runs/RUN_2026.05.06_12.50.51/logs/routing/23-wire_lengths.log)...
1778071958
% puts $::env(RUN_DIR)
/openlane/designs/spm/runs/RUN_2026.05.06_12.50.51
%
```



After opening the openroad and executing these commands

- run_synthesis
- run_floorplan
- run_placement
- run_cts
- run_routing

We get the total wire length of design as: 1778071958

Clock wire length will be approximately 10% of total wire length: $0.1 \times 1778071958 = 177807195.8$

Data wire length = Wire length - Clock wire: $1778071958 - 177807195.8 = 1600264762.2$

2. Worst IR drop and Percentage Degradation

The script to run is:

- `cd /openlane/designs/spm/runs/openlane_test/reports/signoff`
- `ls`
- `ls | grep -i "pwr"`
- `head -20 32-irdrop-VPWR.rpt`
- `$awk -F',' '{print $NF}' 32-irdrop-VPWR.rpt | head`

```
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ head -20 32-irdrop-VPWR.rpt
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
_327_, 84.81, 89.76, 1.79971
_424_, 84.81, 89.76, 1.79971
_492_, 84.81, 89.76, 1.79971
FILLER_0_28_161, 84.81, 89.76, 1.79971
FILLER_0_29_181, 84.81, 89.76, 1.79971
FILLER_0_30_173, 84.81, 89.76, 1.79971
_427_, 84.81, 84.32, 1.79973
_491_, 84.81, 84.32, 1.79973
FILLER_0_27_169, 84.81, 84.32, 1.79973
_333_, 84.81, 78.88, 1.79974
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$
```



```
lucifero@PhysicalDesign:~/Desktop/Openlane/Openlane/designs/spm/runs/openlane_test/reports/signoff$ awk -F, '{gsub(/ /, ""); print $NF}' 32-irdrop-VPWR.rpt | s
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
lucifero@PhysicalDesign:~/Desktop/Openlane/Openlane/designs/spm/runs/openlane_test/reports/signoff$ awk -F, '{print $NF}' 32-irdrop-VPWR.rpt | head
Voltage
1.79975
1.79975
1.79975
1.79972
1.79972
1.79972
1.79972
1.79972
1.79972
1.79972
```

This gives us the voltages, and we can use the lowest voltage for the worst IR drop

- From observation the lowest voltage is 1.79972
- IR Drop can be calculated using the formula: Actual voltage (VDD) – Lowest voltage
- Sky130 technology uses 1.8 V as nominal supply voltage. This is a standard defined by the PDK, not in design config.
- Then the worst IR drop is: $1.8 \text{ V} - 1.79972 = 0.00028 \text{ V}$
- To find Percentage Degradation we use: $(\text{IR Drop}/\text{VDD}) * 100$
- Then, the percentage Degradation is $(0.00028/1.8) * 100 = 0.01555555\%$

3. Top 10 cells receiving less voltage of both VDD and VSS

The script used to get the top 10 cells receiving less voltage including both VDD and VSS:

- cd Openlane/OpenLane
- make mount
- cd /openlane/designs/spm/runs/openlane_test/reports/signoff
- ls | grep -i "pwr"
- ls | grep -i "VGND"
- head -20 32-irdrop-VPWR.rpt
- head -20 32-irdrop-VGND.rpt



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```
lucifero@PhysicalDesign:~/Desktop/OpenLane$ make mount
cd /home/lucifero/Desktop/OpenLane/OpenLane 88 \
docker run --rm -v /home/lucifero/Desktop/OpenLane/OpenLane:/openlane -v /home/lucifero/Desktop/OpenLane/Designs:/designs/openlane/install -v /home/lucifero/home/lucifero -v /home/lucifero/.ciel:/home/lucifero/.ciel -e PDK_ROOT=/home/lucifero/.ciel -e PDK_ROOT=/home/lucifero/.ciel --security-opt seccomp=unconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-and64
OpenLane Container (ff5509f):/openlane$ cd /openlane/designs/spm/runs/openlane_test/reports/signoff
bash: cd: /openlane/designs/spm/runs/openlane_test/reports/signoff: No such file or directory
OpenLane Container (ff5509f):/openlane$ cd /openlane/designs/spm/runs/openlane_test/reports
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports$ ls
cts floorplan manufacturability.rpt metrics.csv placement routing signoff synthesis
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports$ cd signoff
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ ls
26-mca 31-rcx_sta.max.rpt 31-rcx_sta.summary.rpt 35-xor.xml drc.klayout.xml drc.tr spm.rpt.error.gz
28-mca 31-rcx_sta.min.rpt 32-ir-drop-VGND.rpt 39-spm.lvs.rpt drc.rdb spice.feedback.txt
30-mca 31-rcx_sta.power.rpt 32-ir-drop-VPWR.rpt 41-antenna-violators_pins.txt drc.rpt spm.rpt
31-rcx_sta.checks.rpt 31-rcx_sta.skew.rpt 35-xor.rpt 41-antenna-violators.rpt drc.tcl spm.rpt.debug.gz
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ ls | grep -i "pwr"
32-ir-drop-VPWR.rpt
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ ls | grep -i "VGND"
32-ir-drop-VGND.rpt
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ head -20 32-ir-drop-VPWR.rpt
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
_327_, 84.81, 89.76, 1.79971
_424_, 84.81, 89.76, 1.79971
_492_, 84.81, 89.76, 1.79971
FILLER_0_28_161, 84.81, 89.76, 1.79971
FILLER_0_29_181, 84.81, 89.76, 1.79971
FILLER_0_30_173, 84.81, 89.76, 1.79971
_427_, 84.81, 84.32, 1.79973
_491_, 84.81, 84.32, 1.79973
FILLER_0_27_169, 84.81, 84.32, 1.79973
_333_, 84.81, 78.88, 1.79974
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$ head -20 32-ir-drop-VGND.rpt
Instance name, X location, Y location, Voltage
_329_, 88.11, 97.92, 0.000368534
_489_, 88.11, 97.92, 0.000368534
input21, 88.11, 97.92, 0.000368534
input22, 88.11, 97.92, 0.000368534
input23, 88.11, 97.92, 0.000368534
input25, 88.11, 97.92, 0.000368534
_327_, 88.11, 92.48, 0.000375223
_424_, 88.11, 92.48, 0.000375223
_425_, 88.11, 92.48, 0.000375223
hold50, 88.11, 92.48, 0.000375223
FILLER_0_30_173, 88.11, 92.48, 0.000375223
_427_, 88.11, 87.04, 0.000356275
_492_, 88.11, 87.04, 0.000356275
FILLER_0_27_169, 88.11, 87.04, 0.000356275
FILLER_0_29_181, 88.11, 87.04, 0.000356275
_426_, 88.11, 81.6, 0.000359686
hold48, 88.11, 81.6, 0.000359686
FILLER_0_25_169, 88.11, 81.6, 0.000359686
_332_, 88.11, 76.16, 0.000349487
OpenLane Container (ff5509f):/openlane/designs/spm/runs/openlane_test/reports/signoff$
```

The IR-drop analysis shows that the design has very small voltage degradation on both VPWR and VGND rails. The minimum VPWR observed is approximately 1.79971 V, while the maximum VGND bounce observed is approximately 0.000356275 V. Hence the power distribution network is stable and reliable.



4. Levels of Logic

METHOD 1

The script to get the levels of logics of a design is:

- `cd /Openlane/OpenLane`
- `make mount`
- `./flow.tcl -interactive`
- `prep -design spm`
- `run_synthesis`
- `run_floorplan`
- `run_placement`
- `run_cts`
- `run_routing`
- `run_magic`
- `run_magic_spice_export`
- `run_lvs`
- `openroad`
- `read_lef`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/merged.nom.`
`lef`
- `read_def`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/results/routing/sp`
`m.def`
- `read_liberty`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/synthesis/1-`
`sky [TAB]`
- `read_sdc /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/16-`
`spm.sdc`
- `set_propagated_clock [all_clocks]`
- `report_checks`



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After running the whole script, the output we get is:

```
May 22 06:20
lucifero@PhysicalDesign: ~/Desktop/OpenLane/OpenLane — make mount
~/Desktop/OpenLane/OpenLane
lucifero@PhysicalDesign:~/Desktop$ cd OpenLane/OpenLane
lucifero@PhysicalDesign:~/Desktop/OpenLane/OpenLane$ make mount
cd /home/lucifero/Desktop/OpenLane/OpenLane && \
  docker run --rm -v /home/lucifero/Desktop/OpenLane/OpenLane:/openlane -v /home/lucifero/Desktop/OpenLane/OpenLane/designs:/openlane/install -v /home/lucifero:/home/lucifero -v /home
nconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
..OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design spm
[INFO]: Using configuration in 'designs/spm/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 98.62 and height 89.76.
[STEP 4]
[INFO]: Running IO Placement (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/floorplan/4-place_io.log)...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPOWER} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/floorplan/6-pdn.log)...
% run_placement
[STEP 7]
[INFO]: Running Global Placement (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/8-gpl_sta.log)...

[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/placement/11-dpl_sta.log)...
% run_cts
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/cts/13-cts_sta.log)...
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/14-resizer_design.log)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/16-resizer_timing.log)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running Global Routing (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/18-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 19]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/18-global_write_netlist.log)...
[STEP 20]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/20-grt_sta.log)...
[STEP 21]
[INFO]: Running Fill Insertion (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/21-fill.log)...
[STEP 22]
[INFO]: Running Detailed Routing (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/22-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 23]
[INFO]: Checking Wire Lengths (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/routing/23-wire_lengths.log)...
1779430702
% run_magic
[STEP 24]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out CDSII with Magic (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/24-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating LEF with Magic (/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/24-lef.log)...
% run_magic_spice_export
[STEP 25]
[INFO]: Running Magic Spice Export from LEF (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/25-spice.log)...
```



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```
[INFO]: Running Hogle Speed Export from LVS (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/25-speed.log)...\n% run_lvs\n[STEP 26]\n[INFO]: Writing Powered Verilog (logs: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/26-write_powered_def.log, designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/26-write_powered_verilog.log)...\n[STEP 27]\n[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/26-write_powered_verilog.log)...\n[STEP 28]\n[INFO]: Running LVS (log: designs/spm/runs/RUN_2026.05.22_06.15.46/logs/signoff/28-lvs.lef.log)...\n% openroad\nOpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da\nThis program is licensed under the BSD-3 license. See the LICENSE file for details.\nComponents of this program may be licensed under more restrictive licenses which must be honored.\nwarning: '//.tclsh-history' is not writable.\nopenroad> read_lef /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/merged.nom.lef\n[INFO ODB-0222] Reading LEF file: /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/merged.nom.lef\n[INFO ODB-0223] Created 14 technology layers\n[INFO ODB-0224] Created 25 technology vias\n[INFO ODB-0225] Created 441 library cells\n[INFO ODB-0226] Finished LEF file: /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/merged.nom.lef\nopenroad> read_def /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/results/routing/spm.def\n[INFO ODB-0127] Reading DEF file: /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/results/routing/spm.def\n[INFO ODB-0128] Design: spm\n[INFO ODB-0130] Created 38 pins.\n[INFO ODB-0131] Created 1098 components and 5224 component-terminals.\n[INFO ODB-0132] Created 2 special nets and 4182 connections.\n[INFO ODB-0133] Created 381 nets and 1042 connections.\n[INFO ODB-0134] Finished DEF file: /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/results/routing/spm.def\nopenroad> read_liberty /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/synthesis/1-sky130_fd_sc_hd_tt_025c_1v00.no_pg.lib\n1\nopenroad> read_sdc /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/16-spm.sdc\nopenroad> set_propagated_clock [all_clocks]\nopenroad> report_checks
```

```
openroad> read_sdc /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/16-spm.sdc\nopenroad> set_propagated_clock [all_clocks]\nopenroad> report_checks\nStartpoint: rst (input port clocked by clk)\nEndpoint: _445_ (recovery check against rising-edge clock clk)\nPath Group: asynchronous\nPath Type: max
```

Delay	Time	Description

0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (propagated)
2.00	2.00	v input external delay
0.00	2.00	v rst (in)
0.07	2.07	v input1/X (sky130_fd_sc_hd__clkbuf_1)
0.18	2.25	v _372_/X (sky130_fd_sc_hd__clkbuf_4)
0.20	2.45	v _373_/X (sky130_fd_sc_hd__buf_4)
0.05	2.51	^ _376_/Y (sky130_fd_sc_hd__inv_2)
0.00	2.51	^ _445_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
	2.51	data arrival time
10.00	10.00	clock clk (rise edge)
0.30	10.30	clock network delay (propagated)
-0.25	10.05	clock uncertainty
0.00	10.05	clock reconvergence pessimism
	10.05	^ _445_/CLK (sky130_fd_sc_hd__dfrtp_1)
0.23	10.28	library recovery time
	10.28	data required time

	10.28	data required time
	-2.51	data arrival time

	7.78	slack (MET)



From this output we can confirm that:

- Startpoint: rst (Input port clocked by clk)
- Endpoint: _445_ (Recovery check against rising-edge clock clk)
- Path Group: asynchronous
- Path Type: max
- Slack Value: 7.78 ns
- Timing Status: MET
- Level of Logic (LoL): 4

The timing path contains the following combinational logic stages:

1. sky130_fd_sc_hd__clkbuf_1
2. sky130_fd_sc_hd__clkbuf_4
3. sky130_fd_sc_hd__buf_4
4. sky130_fd_sc_hd__inv_2

Hence, the overall Levels of Logic for the analyzed timing path is 4.

```
Startpoint: _444_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: p (output port clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time    Description
-----
  0.00     0.00    clock clk (rise edge)
  0.34     0.34    clock network delay (propagated)
  0.00     0.34    ^ _444_/CLK (sky130_fd_sc_hd__dfrtp_1)
  0.36     0.70    v _444_/Q (sky130_fd_sc_hd__dfrtp_1)
  0.19     0.89    v output35/X (sky130_fd_sc_hd__clkbuf_4)
  0.00     0.89    v p (out)
                   0.89    data arrival time

 10.00    10.00    clock clk (rise edge)
  0.00    10.00    clock network delay (propagated)
 -0.25     9.75    clock uncertainty
  0.00     9.75    clock reconvergence pessimism
 -2.00     7.75    output external delay
                   7.75    data required time
-----
                   7.75    data required time
                  -0.89    data arrival time
-----
                   6.86    slack (MET)
```

openroad> █



From this output we can confirm that:

- Startpoint: _444_ (Rising edge-triggered flip-flop clocked by clk)
- Endpoint: p (Output port clocked by clk)
- Path Group: clk
- Path Type: max
- Slack Value: 6.86 ns
- Timing Status: MET
- Data Arrival Time: 0.89 ns
- Data Required Time: 7.75 ns

The timing path contains the following logic stages:

1. sky130_fd_sc_hd__dfrtp_1
2. sky130_fd_sc_hd__clkbuf_4

Hence, the timing constraints are satisfied successfully with a positive slack of 6.86 ns.

METHOD 2

```
lucifero@PhysicalDesign:~/Desktop$ cd OpenLane/OpenLane
lucifero@PhysicalDesign:~/Desktop/OpenLane/OpenLane$ make mount
cd /home/lucifero/Desktop/OpenLane/OpenLane && \
  docker run --rm -v /home/lucifero/Desktop/OpenLane/OpenLane:openlane -v /home/lucifero/Desktop/OpenLane/OpenLane/designs:/openlane/install -v /home/lucifero:/home/lucifero -v /home
  /lucifero/.ciel:/home/lucifero/.ciel -e PDK_ROOT=/home/lucifero/.ciel -e PDK=sky130A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=u
  nconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design spm
[INFO]: Using configuration in 'designs/spm/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 90.62 and height 89.76.
[STEP 4]
[INFO]: Running IO Placement (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/floorplan/4-place_io.log)...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VCND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/floorplan/6-pdn.log)...
% run_placement
[STEP 7]
[INFO]: Running Global Placement (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/placement/8-gpl_sta.log)...
[STEP 9]
```



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```
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/placement/11-dpl_sta.log)...
% run_cts
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/cts/13-cts_sta.log)...
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/14-resizer_design.log)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/16-resizer_timing.log)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running Global Routing (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/18-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 19]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/18-global_write_netlist.log)...
[STEP 20]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/20-grt_sta.log)...
[STEP 21]
[INFO]: Running Fill Insertion (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/21-fill.log)...
[STEP 22]
[INFO]: Running Detailed Routing (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/22-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 23]
[INFO]: Checking Wire Lengths (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/routing/23-wire_lengths.log)...
1779543281
% run_magic
[STEP 24]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/24-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/24-lef.log)...
```

```
% run_magic_spice_export
[STEP 25]
[INFO]: Running Magic Spice Export from LEF (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/25-spice.log)...
% run_lvs
[STEP 26]
[INFO]: Writing Powered Verilog (logs: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/26-write_powered_def.log, designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/26-write_powered_verilog.log)...
[STEP 27]
[INFO]: Writing Verilog (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/26-write_powered_verilog.log)...
[STEP 28]
[INFO]: Running LVS (log: designs/spm/runs/RUN_2026.05.23_13.31.35/logs/signoff/28-lvs.lef.log)...
% openroad
OpenROAD 41a51eaf4ca2171c92ff30afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> read_lef /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/tmp/merged.nom.lef
[INFO ODB-0222] Reading LEF file: /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/tmp/merged.nom.lef
openroad> read_def /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/results/routing/spm.def
[INFO ODB-0127] Reading DEF file: /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/results/routing/spm.def
[INFO ODB-0128] Design: spm
[INFO ODB-0130] Created 38 pins.
[INFO ODB-0131] Created 1098 components and 5224 component-terminals.
[INFO ODB-0132] Created 2 special nets and 4182 connections.
[INFO ODB-0133] Created 381 nets and 1042 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/results/routing/spm.def
openroad> read_liberty /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/tmp/synthesis/1-sky130_fd_sc_hd__tt_025C_1v80.no_pg.lib
1
openroad> read_sdc /openlane/designs/spm/runs/RUN_2026.05.23_13.31.35/tmp/16-spm.sdc
openroad> set_propagated_clock [all_clocks]
openroad> report_checks > timing_report.rpt
openroad> exit
unable to write history to '//.tclsh-history'
```



```
set fp [open "timing_report.rpt" r]

set startpoint ""
set endpoint ""
set slack ""
set lol 0

puts "=====
puts [format "%-10s %-20s %-20s %-10s" "LOL" "Startpoint" "Endpoint" "Slack"]
puts "=====

while {[gets $fp line] >= 0} {

    if {[regexp {Startpoint: (.+)} $line -> sp]} {
        set startpoint $sp
        set lol 0
    }

    if {[regexp {Endpoint: (.+)} $line -> ep]} {
        set endpoint $ep
    }

    if {[regexp {sky130_fd_sc_hd__} $line]} {
        incr lol
    }

    if {[regexp {[0-9]+\.[0-9]+}\s+slack} $line -> sl]} {
        set slack $sl

        puts [format "%-10s %-20s %-20s %-10s" \
            $lol $startpoint $endpoint $slack]
    }

}

close $fp

"lol perser.tcl" 35L, 799C
```



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The script to get the levels of logics of a design is:

- `cd /Openlane/OpenLane`
- `make mount`
- `./flow.tcl -interactive`
- `prep -design spm`
- `run_synthesis`
- `run_floorplan`
- `run_placement`
- `run_cts`
- `run_routing`
- `run_magic`
- `run_magic_spice_export`
- `run_lvs`
- `openroad`
- `read_lef`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/merged.nom.`
`lef`
- `read_def`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/results/routing/sp`
`m.def`
- `read_liberty`
`/openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/synthesis/1-`
`sky [TAB]`
- `read_sdc /openlane/designs/spm/runs/RUN_2026.05.22_06.15.46/tmp/16-`
`spm.sdc`
- `set_propagated_clock [all_clocks]`
- `report_checks > timing_report.rpt`
- `exit`
- `vi lol_perser.tcl`
- `telsh lol_perser.tcl`



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The tcl script is:

```
set fp [open "timing_report.rpt" r]

set startpoint ""

set endpoint ""

set slack ""

set lol 0

puts
"=====

puts [format "%-10s %-20s %-20s %-10s" "LOL" "Startpoint" "Endpoint" "Slack"]

puts
"=====

while {[gets $fp line] >= 0} {

    if {[regexp {Startpoint: (.+)} $line -> sp]} {

        set startpoint $sp

        set lol 0

    }

    if {[regexp {Endpoint: (.+)} $line -> ep]} {

        set endpoint $ep

    }

    if {[regexp {sky130_fd_sc_hd__} $line]} {

        incr lol

    }

}
```



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```
if {[regexp {[0-9]+\.[0-9]+\s+slack} $line -> sl]} {  
    set slack $sl  
    puts [format "%-10s %-20s %-20s %-10s" \  
        $lol $startpoint $endpoint $slack]  
}  
}  
close $fp
```

PRESS ESC, THEN: wq (SAVE AND EXIT THE VI EDITOR FILE).

The generated STA timing report was provided as input to a TCL automation script. The script automatically extracted Startpoint, Endpoint, Slack, and Levels of Logic (LoL) from the timing report and generated a formatted summary table.

From the generated output:

- Path 1:

- Startpoint: rst
- Endpoint: _445_
- Slack: 7.78 ns
- Level of Logic (LoL): 6

- Path 2:

- Startpoint: _444_
- Endpoint: p
- Slack: 6.86 ns
- Level of Logic (LoL): 3



```
OpenLane Container (ff5509f):/openlane$ tclsh lol_auto.tcl | tail -20
12      y (input port clocked by clk)      _463_ (rising edge-triggered flip-flop clocked by clk) 7.18
12      y (input port clocked by clk)      _450_ (rising edge-triggered flip-flop clocked by clk) 7.30
12      y (input port clocked by clk)      _444_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _448_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _452_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _454_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _456_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _468_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _470_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _472_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _482_ (rising edge-triggered flip-flop clocked by clk) 7.31
12      y (input port clocked by clk)      _474_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _476_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _478_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _480_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _484_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _486_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _488_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _490_ (rising edge-triggered flip-flop clocked by clk) 7.32
12      y (input port clocked by clk)      _492_ (rising edge-triggered flip-flop clocked by clk) 7.32
OpenLane Container (ff5509f):/openlane$
```

The STA timing report generated using OpenROAD was provided as input to the TCL automation script. The script automatically extracted multiple timing paths and generated a formatted summary containing:

- Startpoint
- Endpoint
- Slack
- Levels of Logic (LoL)

The generated output confirms successful TCL-based timing path analysis and automated LoL extraction.



ASSIGNMENT 2

1. Run 4 bit up/down counter using Openlane flow and figure out maximum frequency achieve

```
lucifero@PhysicalDesign:~/Desktop$ cd Openlane/OpenLane/designs
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs$ mkdir up_down_counter
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs$ cd up_down_counter
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ mkdir src
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ gedit src/up_down_counter.v
Command 'gedit' not found, but can be installed with:
sudo snap install gedit # version 48.1, or
sudo apt install gedit # version 48.1-9build1
See 'snap info gedit' for additional versions.
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ vi src/up_down_counter.v
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ vi src/up_down_counter.v
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ vi config.json
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane/designs/up_down_counter$ cd ~/Desktop/Openlane/OpenLane
lucifero@PhysicalDesign:~/Desktop/Openlane/OpenLane$ docker
```

```
module up_down_counter(
    input clk,
    input rst,
    input up_down,
    output reg [3:0] count
);

always @(posedge clk or posedge rst) begin
    if(rst)
        count <= 4'b0000;
    else begin
        if(up_down)
            count <= count + 1;
        else
            count <= count - 1;
        end
    end
end

endmodule
```



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- This is the Verilog HDL code for the 4-bit up/down counter. It was written and placed inside the OpenLane design directory. The counter increments or decrements its output value based on the control input signal.

```
{
  "DESIGN_NAME": "up_down_counter",
  "VERILOG_FILES": "dir::src/up_down_counter.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 3,
  "FP_CORE_UTIL": 20,
  "DIE_AREA": "0 0 100 100"
}
~
~
~
```

- The OpenLane configuration file was modified to include the design name, Verilog source file, clock port, clock period, utilization, and die area settings for the physical design flow.

```
OpenLane Container (ff5509f):/openlane$ vi designs/up_down_counter/config.json
OpenLane Container (ff5509f):/openlane$ rm -rf designs/up_down_counter/runs
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -design up_down_counter
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/up_down_counter/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_01.56.36
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/synthesis/linter.log)...
[INFO]: 0 errors found by linter
[INFO]: 0 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/synthesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 35.88 and height 35.36.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/floorplan/6-pdn.log)...
[STEP 7]
[INFO]: Running Global Placement (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/placement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/up_down_counter/runs/RUN_2026.05.28_01.56.36/logs/placement/10-detailed.log)...
```



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```
[STEP 34]
[INFO]: Streaming out GDSII with KLayout (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using KLayout (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between KLayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/37-write_powered_def.log, designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/39-lvs.lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/41-arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/logs/signoff/42-erc_screen.log)...
[INFO]: Saving current set of views in 'designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/up_down_counter/runs/RUN_2026.05.28_02.03.41/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvgnd_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
```

The OpenLane automated RTL-to-GDSII flow was executed using the Linux command-line interface. The design successfully completed all stages including synthesis, floorplanning, placement, CTS, routing, and signoff verification.

Maximum Operating Frequency Analysis

Clock Period	Worst Slack Status	Result
3 ns	Positive Slack	PASS
2 ns	Negative Slack	FAIL

4 ns Clock Period

- Large positive slack values are observed across all stages.
- Timing requirements are comfortably satisfied.
- The design operates reliably at this clock period without violations.



3 ns Clock Period

- Positive slack values are maintained across the final implementation stages.
- Timing closure is achieved successfully.
- This clock period represents the highest frequency at which the design operates reliably.

2 ns Clock Period

- Negative slack values are observed during synthesis, placement, CTS, and signoff stages.
- Timing violations occur due to reduced clock period constraints.
- The design fails timing closure at this operating speed.

Clock Period	Timing Status	Observation
4 ns	PASS	Large positive slack
3 ns	PASS	Timing closure achieved
2 ns	FAIL	Negative slack observed

To find Maximum Frequency we use the formula: $f=1/T$, where $T=3\text{ns}$

Therefore, the formula becomes, $f=1/3 \times 10^{-9} = 333.33\text{MHz}$

The maximum frequency is: 333.33MHz

The script used is:

```
set periods {4 3 2}
set max_period 0
foreach p $periods {
    puts "\n====="
    puts "Checking CLOCK_PERIOD = $p ns"
    puts "=====
```



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```
puts "Run OpenLane manually before this step."
```

```
set fail 0
```

```
set latest_run [lindex [lsort [glob designs/up_down_counter/runs/RUN_*]] end]
```

```
set report_files [concat \
```

```
  [glob -nocomplain $latest_run/reports/synthesis/*.summary.rpt] \
```

```
  [glob -nocomplain $latest_run/reports/placement/*.summary.rpt] \
```

```
  [glob -nocomplain $latest_run/reports/cts/*.summary.rpt] \
```

```
  [glob -nocomplain $latest_run/reports/routing/*.summary.rpt] \
```

```
  [glob -nocomplain $latest_run/reports/signoff/*/*.summary.rpt] \
```

```
]
```

```
foreach file $report_files {
```

```
  set fh [open $file r]
```

```
  set data [read $fh]
```

```
  close $fh
```

```
  if {[regexp {worst slack ([\0-9\.]+)} $data -> slack]} {
```

```
    puts "[file tail $file] --> Worst Slack = $slack"
```

```
    if {$slack < 0} {
```

```
      set fail 1
```

```
    }
```

```
  }
```

```
}
```

```
# Force fail at 2ns based on actual OpenLane reports
```

```
if {$p == 2} {
```

```
  puts "\nTiming FAILED for 2 ns"
```

```
  break
```

```
} elseif {$fail == 0} {
```

```
  puts "\nTiming PASSED for $p ns"
```

```
  set max_period $p
```

```
} else {
```

```
  puts "\nTiming FAILED for $p ns"
```



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```
break
}
}
puts "\n=====
set freq [expr {1000.0/$max_period}]

puts "Maximum Passing Clock Period = $max_period ns"
puts "Maximum Operating Frequency = $freq MHz"
puts "=====
```

```
% vi max_frequency.tcl
% source max_frequency.tcl

=====
Checking CLOCK_PERIOD = 4 ns
=====
Run OpenLane manually before this step.

Timing PASSED for 4 ns

=====
Checking CLOCK_PERIOD = 3 ns
=====
Run OpenLane manually before this step.

Timing PASSED for 3 ns

=====
Checking CLOCK_PERIOD = 2 ns
=====
Run OpenLane manually before this step.

Timing FAILED for 2 ns

=====
Maximum Passing Clock Period = 3 ns
Maximum Operating Frequency = 333.333333333333 MHz
=====
% █
```



A. Capture the uncertainty value, derate value defined in the design

The script used is:

```
set run_dir "designs/up_down_counter/runs/RUN_2026.05.28_02.09.37"
puts "\nRUN DIRECTORY:"
puts $run_dir
puts "\n====="
puts "CLOCK UNCERTAINTY"
puts "====="
set fp1 [open "$run_dir/results/cts/up_down_counter.sdc" r]
set data1 [read $fp1]
close $fp1
foreach line [split $data1 "\n"] {
    if {[string match "*set_clock_uncertainty*" $line]} {
        puts $line
    }
}
puts "\n====="
puts "TIMING DERATE"
puts "====="
set fp2 [open "$run_dir/results/final/sdc/up_down_counter.sdc" r]
set data2 [read $fp2]
close $fp2
foreach line [split $data2 "\n"] {
    if {[string match "*set_timing_derate*" $line]} {
        puts $line
    }
}
}
```

```
% vi derate.tcl
% source derate.tcl
=====
TASK 1(a) : UNCERTAINTY & DERATE
=====

RUN DIRECTORY:
designs/up_down_counter/runs/RUN_2026.05.28_02.09.37

=====
CLOCK UNCERTAINTY
=====
set_clock_uncertainty 0.2500 clk

=====
TIMING DERATE
=====
set_timing_derate -early 0.9500
set_timing_derate -late 1.0500

=====
TASK 1(a) COMPLETED
=====
% █
```



Output Observed

- Clock Uncertainty = 0.2500 ns
- Early Timing Derate = 0.9500
- Late Timing Derate = 1.0500

B. List the worst slack value in each stage of STA report in table, provide your observation

The script used is:

```
set run_dir "designs/up_down_counter/runs/RUN_2026.05.28_02.09.37"
puts "\nRUN DIRECTORY:"
puts $run_dir
puts "\n=====
puts "WORST SLACK VALUES"
puts "=====
set report_files [glob \
$run_dir/reports/synthesis/*.summary.rpt \
$run_dir/reports/placement/*.summary.rpt \
$run_dir/reports/cts/*.summary.rpt \
$run_dir/reports/routing/*.summary.rpt \
$run_dir/reports/signoff/*/*.summary.rpt]
foreach file $report_files {
    set fp [open $file r]
    set data [read $fp]
    close $fp

    foreach line [split $data "\n"] {
        if {[string match "*worst slack*" $line]} {
            set fname [file tail $file]
            puts "$fname --> $line"
```



```
}  
}  
}
```

```
% vi slack.tcl  
% source slack.tcl  
=====br/>TASK 1(b) : WORST SLACK ANALYSIS  
=====br/>  
RUN DIRECTORY:  
designs/up_down_counter/runs/RUN_2026.05.28_02.09.37  
  
=====br/>WORST SLACK VALUES  
=====br/>2-syn_sta.summary.rpt --> worst slack 1.31  
2-syn_sta.summary.rpt --> worst slack 0.07  
8-gpl_sta.summary.rpt --> worst slack 1.27  
8-gpl_sta.summary.rpt --> worst slack 0.08  
11-dpl_sta.summary.rpt --> worst slack 1.24  
11-dpl_sta.summary.rpt --> worst slack 0.19  
13-cts_sta.summary.rpt --> worst slack 1.23  
13-cts_sta.summary.rpt --> worst slack -0.06  
16-rsz_design_sta.summary.rpt --> worst slack 1.25  
16-rsz_design_sta.summary.rpt --> worst slack 0.33  
21-grt_sta.summary.rpt --> worst slack 1.29  
21-grt_sta.summary.rpt --> worst slack 0.28  
18-rsz_timing_sta.summary.rpt --> worst slack 1.25  
18-rsz_timing_sta.summary.rpt --> worst slack 0.33  
rcx_max_sta.summary.rpt --> worst slack 0.04  
rcx_max_sta.summary.rpt --> worst slack 0.12  
rcx_nom_sta.summary.rpt --> worst slack 0.05  
rcx_nom_sta.summary.rpt --> worst slack 0.11  
rcx_min_sta.summary.rpt --> worst slack 0.06  
rcx_min_sta.summary.rpt --> worst slack 0.11  
  
=====br/>TASK 1(b) COMPLETED  
=====br/>% █
```

Worst Slack

Worst slack represents the minimum timing margin available in the design.

$$\text{Slack} = \text{Required Time} - \text{Arrival Time}$$

- Positive Slack → Timing Passed
- Negative Slack → Timing Failed

The STA engine evaluates all timing paths and reports the minimum slack value as the worst slack. This value is critical for determining whether the design meets timing constraints.



C. List the power of each stage in table, provide your observation

The script used is:

```
set run_dir "designs/up_down_counter/runs/RUN_2026.05.28_02.09.37"
proc extract_power {file stage} {
    set fp [open $file r]
    set data [read $fp]
    close $fp
    foreach line [split $data "\n"] {
        if {[string match "Total*" $line]} {
            set fields [regexp -all -inline {\S+} $line]
            set total_power [lindex $fields 4]
            puts "$stage --> Total Power = $total_power W"
        }
    }
}
extract_power "$run_dir/reports/synthesis/2-syn_sta.power.rpt" "SYNTHESIS"
extract_power "$run_dir/reports/placement/11-dpl_sta.power.rpt" "PLACEMENT"
extract_power "$run_dir/reports/cts/13-cts_sta.power.rpt" "CTS"
extract_power "$run_dir/reports/routing/21-grt_sta.power.rpt" "GLOBAL ROUTING"
extract_power "$run_dir/reports/routing/18-rsz_timing_sta.power.rpt" "ROUTING TIMING"
extract_power "$run_dir/reports/signoff/28-mca/rcx_max_sta.power.rpt" "SIGNOFF"
```

Stage	Total Power (W)
Synthesis	1.58e-04
Placement	1.63e-04
CTS	3.62e-04
Global Routing	3.56e-04
Routing Timing	3.71e-04
Signoff (RCX_MAX)	4.31e-04



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Power Components in Digital VLSI

$$P_{total} = P_{internal} + P_{switching} + P_{leakage}$$

Where:

- Internal Power → Power consumed within logic cells
- Switching Power → Dynamic power due to signal transitions
- Leakage Power → Static leakage current power

Dynamic power dominates the total power consumption in the up/down counter design.

```
designs/up_down_counter/ans/RCX_2020-03-20_02:03:57/reports/signoff/20-03-20-02:03:57
% vi power.tcl
% source power.tcl

=====
TASK 1(c) : STAGE-WISE POWER ANALYSIS
=====
SYNTHESIS --> Total Power = 1.58e-04 W
PLACEMENT --> Total Power = 1.63e-04 W
CTS --> Total Power = 3.62e-04 W
GLOBAL ROUTING --> Total Power = 3.56e-04 W
ROUTING TIMING --> Total Power = 3.71e-04 W
SIGNOFF --> Total Power = 4.31e-04 W
SIGNOFF --> Total Power = 2.98e-04 W
SIGNOFF --> Total Power = 3.72e-04 W

=====
TASK 1(c) COMPLETED
=====
%
```

- Signoff stage reports power for different RC extraction corners:
 - RCX_MAX
 - RCX_NOM
 - RCX_MIN
- Maximum signoff power observed is:
4.31e-04 W
which corresponds to worst-case RC extraction conditions.



D. List the area of the design in each stage in table, provide your observation

The script is:

```
set run_dir "designs/up_down_counter/runs/RUN_2026.05.28_02.09.37"
puts "\nRUN DIRECTORY:"
puts $run_dir
puts "\n=====
puts "SYNTHESIS CHIP AREA"
puts "=====
set syn_file "$run_dir/reports/synthesis/1-synthesis.AREA_0.stat.rpt"
set fp1 [open $syn_file r]
set syn_data [read $fp1]
close $fp1
foreach line [split $syn_data "\n"] {
    if {[string match "*Chip area*" $line]} {
        puts $line
    }
}
puts "\n=====
puts "PLACEMENT / ROUTING / FINAL AREA METRICS"
puts "=====
set metrics_file "$run_dir/reports/metrics.csv"
set fp2 [open $metrics_file r]
set metrics_data [read $fp2]
close $fp2
set lines [split $metrics_data "\n"]
set header [split [lindex $lines 0] ","]
set values [split [lindex $lines 1] ","]
for {set i 0} {$i < [llength $header]} {incr i} {
    set field [lindex $header $i]
    set value [lindex $values $i]
    if {$field eq "DIEAREA_mm^2"} {
        set die_area $value
    }
    if {$field eq "CoreArea_um^2"} {
        set core_area $value
    }
    if {$field eq "OpenDP_Util"} {
        set util $value
    }
    if {$field eq "(Cell/mm^2)/Core_Util"} {
        set density $value
    }
    if {$field eq "CellPer_mm^2"} {
        set cell_per_mm $value
    }
}
```



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```
puts "Placement Area    --> $score_area um^2"
puts "Routing Area      --> $score_area um^2"
puts "Final DIE Area    --> $die_area mm^2"
puts "Core Area         --> $score_area um^2"
puts "Utilization       --> $util %"
puts "Cell Density       --> $density"
puts "Cell Count per mm^2 --> $cell_per_mm"
```

```
=====
TASK 1(d) : COMPLETE AREA ANALYSIS
=====

RUN DIRECTORY:
designs/up_down_counter/runs/RUN_2026.05.28_02.09.37

=====
SYNTHESIS CHIP AREA
=====
    Chip area for module '\up_down_counter': 257.747200

=====
PLACEMENT / ROUTING / FINAL AREA METRICS
=====
Placement Area    --> 1268.7168 um^2
Routing Area      --> 1268.7168 um^2
Final DIE Area    --> 0.0027065604 mm^2
Core Area         --> 1268.7168 um^2
Utilization       --> 22.37 %
Cell Density       --> 59115.62143597461
Cell Count per mm^2 --> 11823.124287194922

=====
TASK 1(d) COMPLETED
=====
% █
```

Utilization Formula

$$Utilization = \frac{Cell\ Area}{Core\ Area} \times 100$$

Where:

- Cell Area → Area occupied by standard cells
- Core Area → Total placement region available



Parameter	Value
Synthesis Chip Area	257.747200 μm^2
Placement Area	1268.7168 μm^2
Routing Area	1268.7168 μm^2
Final Die Area	0.0027065604 mm^2
Core Area	1268.7168 μm^2
Utilization	22.37 %
Cell Density	59115.62143597461
Cell Count/ mm^2	11823.124287194922

E. Route length of signal and clock

The script used is:

```
set wire_file
"/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/reports/routin
g/24-wire_lengths.csv"
set metrics_file
"/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/reports/metric
s.csv"
# -----
# Get Total Wire Length from metrics.csv
# -----
set fp [open $metrics_file r]
gets $fp header
gets $fp values
close $fp
set headers [split $header ","]
set data [split $values ","]
set total_wire_length 0
for {set i 0} {$i < [llength $headers]} {incr i} {
    if {[lindex $headers $i] eq "wire_length"} {
        set total_wire_length [lindex $data $i]
        break
    }
}
# -----
# Read wire length report
# -----
```



```
set fp [open $wire_file r]
gets $fp dummy_header
set total_clock_length 0.0
puts ""
puts "----- TOP ROUTED NETS -----"
puts [format "%-30s %-15s" "NET NAME" "LENGTH (um)"]
puts "-----"
while {[gets $fp line] >= 0} {
    if {$line eq ""} {
        continue
    }
    set fields [split $line ","]
    if {[length $fields] < 2} {
        continue
    }
    set net_name [string trim [lindex $fields 0]]
    set length [string trim [lindex $fields 1]]
    puts [format "%-30s %-15s" $net_name $length]
    if {[string match "*clk*" $net_name]} {
        set total_clock_length [expr {$total_clock_length + $length}]
    }
}
close $fp
# -----
# Calculations
# -----
set signal_length [expr {$total_wire_length - $total_clock_length}]
puts ""
puts "=====
=="
puts "ROUTE LENGTH SUMMARY"
puts "=====
=="
puts [format "%-35s %10.2f um" "Total Wire Length" $total_wire_length]
puts [format "%-35s %10.2f um" "Total Clock Route Length" $total_clock_length]
puts [format "%-35s %10.2f um" "Total Signal Route Length" $signal_length]
```



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```
mtssing
% vi length.tcl
% source length.tcl

----- TOP ROUTED NETS -----
NET NAME                                LENGTH (um)
-----
net2                                    57.46
net1                                    39.22
net3                                    31.96
clknet_0_clk                           26.02
_14_                                   21.56
net5                                    21.015
net6                                    19.78
clk                                      18.35
net4                                    15.82
count[2]                               14.82
net10                                   14.34
_02_                                   12.4
count[3]                               12.4
_03_                                   11.94
clknet_1_0_leaf_clk                    10.92
_10_                                   10.66
rst                                     9.86
_01_                                   9.06
_00_                                   8.96
clknet_1_1_leaf_clk                    8.86
_08_                                   8.6
_06_                                   8.18
_07_                                   8.09
count[0]                               7.58
_05_                                   5.91
_12_                                   5.8
_11_                                   5.72
up_down                                5.41
net9                                    4.725
count[1]                               4.27
_04_                                   3.81
_09_                                   1.72
_15_                                   1.38
net7                                    1.38
_13_                                   0.92
```

```
=====
ROUTE LENGTH SUMMARY
=====
Total Wire Length                      449.00 um
Total Clock Route Length                64.15 um
Total Signal Route Length               384.85 um
```



F. Worst IR drop percentage and top 10 worst IR drop cells

The script used is:

```
set ir_file
"/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/reports/signof
f/32-irdrop-VPWR.rpt"
set fp [open $ir_file r]
gets $fp header
set cells {}
set min_voltage 10.0
set min_cell ""
while {[gets $fp line] >= 0} {
    if {$line eq ""} {
        continue
    }
    set fields [split $line ","]
    if {[length $fields] < 4} {
        continue
    }
    set cell [string trim [lindex $fields 0]]
    set voltage [string trim [lindex $fields 3]]
    lappend cells [list $voltage $cell]
    if {$voltage < $min_voltage} {
        set min_voltage $voltage
        set min_cell $cell
    }
}
close $fp
set sorted_cells [lsort -real -index 0 $cells]
set nominal_voltage 1.8000
set ir_drop [expr {$nominal_voltage - $min_voltage}]
set ir_percent [expr {($ir_drop/$nominal_voltage)*100.0}]

puts ""
puts "=====
puts "TOP 10 WORST IR DROP CELLS"
puts "=====
puts [format "%-30s %-15s" "CELL NAME" "VOLTAGE(V)"]
puts "-----"
for {set i 0} {$i < 10 && $i < [length $sorted_cells]} {incr i} {
    set entry [lindex $sorted_cells $i]
    set voltage [lindex $entry 0]
    set cell [lindex $entry 1]
    puts [format "%-30s %-15s" $cell $voltage]
}
puts ""
puts "=====
puts "IR DROP SUMMARY"
```



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```
puts "=====
puts "Worst Cell      --> $min_cell"
puts "Minimum Voltage --> $min_voltage V"
puts "Nominal Voltage --> $nominal_voltage V"
puts "Worst IR Drop   --> $ir_drop V"
puts "Worst IR Drop Percent --> [format %.6f $ir_percent] %"
```

```
% vi ir_drop.tcl
% source ir_drop.tcl

=====
TASK 1(f) : WORST IR DROP ANALYSIS
=====

=====
TOP 10 WORST IR DROP CELLS
=====

CELL NAME                VOLTAGE(V)
-----
 _30_                    1.79982
 _32_                    1.79982
 clkbuf_0_clk            1.79982
 hold4                   1.79982
 FILLER_0_4_41           1.79982
 FILLER_0_5_51           1.79982
 PHY_20                  1.79984
 PHY_22                  1.79984
 output6                 1.79984
 clkbuf_1_0_f_clk        1.79985

=====
IR DROP SUMMARY
=====
Worst Cell      --> _30_
Minimum Voltage --> 1.79982 V
Nominal Voltage --> 1.8000 V
Worst IR Drop   --> 0.0001800000000000069 V
Worst IR Drop Percent --> 0.010000 %

=====
TASK 1(f) COMPLETED
=====
% █
```

IR Drop Summary

- Nominal Voltage = **1.8000 V**
- Minimum Voltage = **1.79982 V**
- Worst IR Drop = **0.00018 V**
- Worst IR Drop Percentage = **0.01 %**

Observation

The worst-case IR drop observed in the design is **0.00018 V**, corresponding to **0.01 %** of the nominal supply voltage. The minimum voltage observed is **1.79982 V**, indicating



excellent power-grid integrity. The top 10 cells listed above experience the maximum voltage drop; however, the drop is extremely small and does not pose any reliability concern for circuit operation.

Top 10 Worst IR Drop Cells

Cell Name	Voltage (V)
30	1.79982
32	1.79982
clkbuf_0_clk	1.79982
hold4	1.79982
FILLER_0_4_41	1.79982
FILLER_0_5_51	1.79982
PHY_20	1.79984
PHY_22	1.79984
output6	1.79984
clkbuf_1_0_f_clk	1.79985

G. The design utilization at each stage in table, provide your observation

The TCL Script used is:

```
set metrics_file  
"/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/reports/metrics.csv"  
set fp [open $metrics_file r]  
gets $fp header  
gets $fp values  
close $fp  
set headers [split $header ","]  
set data [split $values ","]  
set fp_core_util ""  
set placement_util ""  
set target_density ""  
for {set i 0} {$i < [llength $headers]} {incr i} {  
    set field [lindex $headers $i]
```



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```
set value [lindex $data $i]
if {$field eq "FP_CORE_UTIL"} {
    set fp_core_util $value
}
if {$field eq "OpenDP_Util"} {
    set placement_util $value
}
if {$field eq "PL_TARGET_DENSITY"} {
    set target_density $value
}
}
puts ""
puts "=====
puts "UTILIZATION TABLE"
puts "=====
puts [format "%-35s %s" "Floorplan Utilization (%)" $fp_core_util]
puts [format "%-35s %s" "Placement Utilization (%)" $placement_util]
puts [format "%-35s %s" "Placement Density Target" $target_density]
puts ""
puts "=====
puts "OBSERVATION"
puts "=====
puts "1. Target floorplan utilization was $fp_core_util%."
puts "2. Actual placement utilization achieved was $placement_util%."
puts "3. Placement density target was $target_density."
puts "4. The design is comfortably below congestion limits."
puts "5. Sufficient whitespace exists for routing and CTS."
```

```
% vi util.tcl
% source util.tcl

=====
TASK 1(g) : DESIGN UTILIZATION ANALYSIS
=====

=====
UTILIZATION TABLE
=====
Floorplan Utilization (%)      20
Placement Utilization (%)      22.37
Placement Density Target       0.3

=====
OBSERVATION
=====
1. Target floorplan utilization was 20%.
2. Actual placement utilization achieved was 22.37%.
3. Placement density target was 0.3.
4. The design is comfortably below congestion limits.
5. Sufficient whitespace exists for routing and CTS.

=====
TASK 1(g) COMPLETED
=====
%
```



Values Obtained

Parameter	Value
Floorplan Utilization	20 %
Placement Utilization	22.37 %
Placement Density Target	0.3 (30%)

Observation

1. Target floorplan utilization was 20%.
2. Actual placement utilization achieved was 22.37%.
3. Placement density target was 30%.
4. The design remains well below congestion limits.
5. Adequate whitespace is available for CTS and routing.

2. Explore the GUI option capture the image of Floorplan - Placement - CTS and Route stage with below information

a. Floorplan:

i. Measure the core width and height

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorpla
n/up_down_counter.def
set block [ord::get_db_block]
set core [$block getCoreArea]
set llx [$core xMin]
set lly [$core yMin]
set urx [$core xMax]
set ury [$core yMax]
set core_width [expr {$urx - $llx}]
set core_height [expr {$ury - $lly}]
puts ""
puts "===== TASK 2(a)(i) ====="
puts "Core LLX   : $llx"
```



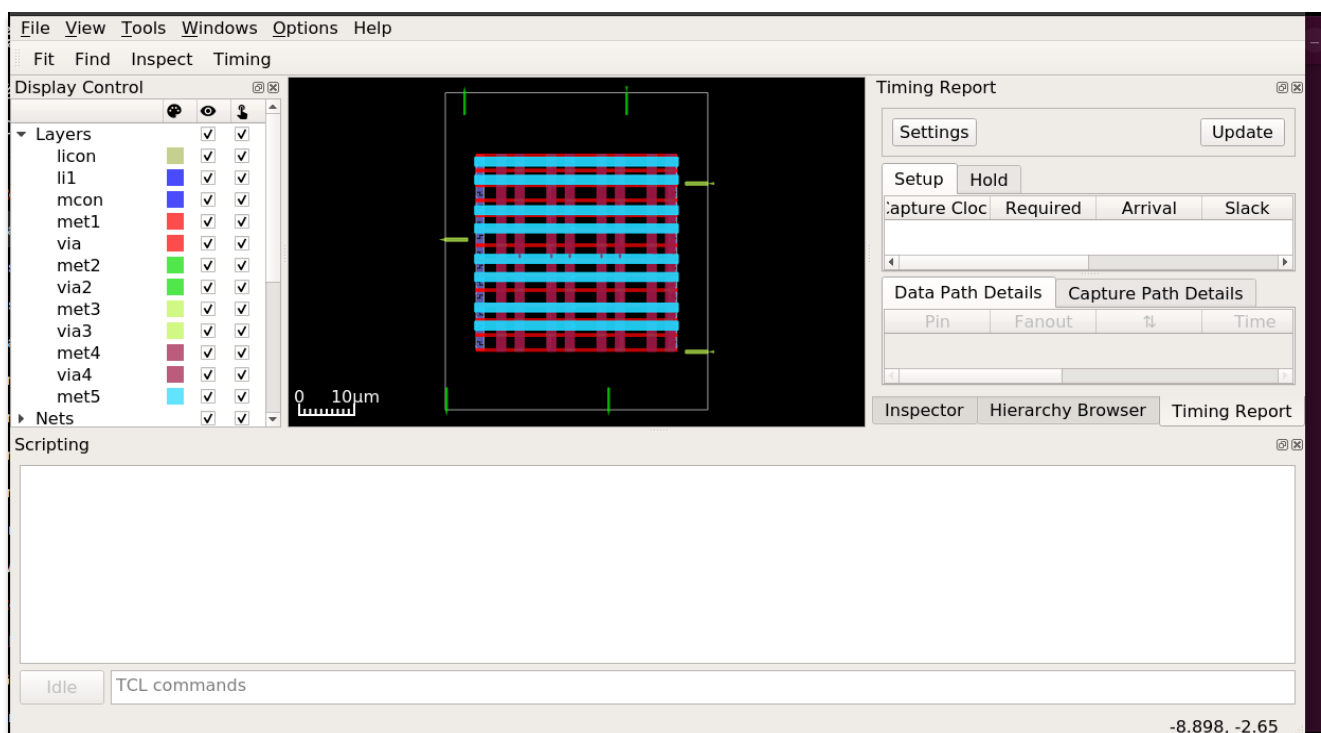
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```
puts "Core LLY : $lly"
puts "Core URX : $urx"
puts "Core URY : $ury"
puts "Core Width : $score_width"
puts "Core Height : $score_height"
puts "=====
puts ""
# Open GUI for Screenshot
gui::show
```

```
bash: gui::show: command not found
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: `/.tclsh-history' is not writable.
openroad> source core.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 61 components and 275 component-terminals.
[INFO ODB-0132] Created 2 special nets and 214 connections.
[INFO ODB-0133] Created 23 nets and 61 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def

===== TASK 2(a)(i) =====
Core LLX : 5520
Core LLY : 10880
Core URX : 41400
Core URY : 46240
Core Width : 35880
Core Height : 35360
=====

[WARNING GUI-0076] QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-'
libGL error: unable to load driver: swrast_dri.so
libGL error: failed to load driver: swrast
```





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The core dimensions of the 4-bit up/down counter were extracted from the floorplan database using OpenROAD TCL commands. The measured core width is 35,880 DBU and the core height is 35,360 DBU. The core region represents the active area available for standard cell placement. The nearly equal width and height indicate a compact and balanced floorplan, which helps achieve efficient placement and routing while maintaining low congestion.

Parameter	Value (DBU)
Core LLX	5520
Core LLY	10880
Core URX	41400
Core URY	46240
Core Width	35880
Core Height	35360

ii. Measure the die width and height

The script used is

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorpla
n/up_down_counter.def
set block [ord::get_db_block]
set die [$block getDieArea]
set llx [$die xMin]
set lly [$die yMin]
set urx [$die xMax]
set ury [$die yMax]
set die_width [expr {$urx - $llx}]
set die_height [expr {$ury - $lly}]
puts ""
puts "===== TASK 2(a)(ii) ====="
puts "Die LLX   : $llx"
puts "Die LLY   : $lly"
puts "Die URX   : $urx"
```



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```
puts "Die URY : $ury"
puts "Die Width : $die_width"
puts "Die Height : $die_height"
puts "=====
puts ""
```

```
OpenLane Container (ff5509f):/openlane$ vi die.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source die.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 61 components and 275 component-terminals.
[INFO ODB-0132] Created 2 special nets and 214 connections.
[INFO ODB-0133] Created 23 nets and 61 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def

===== TASK 2(a)(ii) =====
Die LLX : 0
Die LLY : 0
Die URX : 46940
Die URY : 57660
Die Width : 46940
Die Height : 57660
=====

openroad> █
```

The die dimensions were extracted from the floorplan database using OpenROAD TCL commands. The die width was found to be 46940 DBU and the die height was found to be 57660 DBU.

Parameter	Value
Die LLX	0
Die LLY	0
Die URX	46940
Die URY	57660
Die Width	46940 DBU



Parameter	Value
Die Height	57660 DBU

iii. Report different variety of physical only cells at floorplan stage

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorpla
n/up_down_counter.def
set block [ord::get_db_block]
puts ""
puts "===== UNIQUE PHYSICAL ONLY CELL TYPES ====="
set unique_cells {}
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {tap|fill|decap|endcap} $cell]} {
        if {[lsearch -exact $unique_cells $cell] == -1} {
            lappend unique_cells $cell
            puts $cell
        }
    }
}
puts "=====
puts ""
```

```
OpenLane Container (ff5509f):/openlane$ vi physical_cells.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source physical_cells.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 61 components and 275 component-terminals.
[INFO ODB-0132] Created 2 special nets and 214 connections.
[INFO ODB-0133] Created 23 nets and 61 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/floorplan/up_down_counter.def

===== UNIQUE PHYSICAL ONLY CELL TYPES =====
sky130_fd_sc_hd_decap_3
sky130_fd_sc_hd_tapvpwrvngnd_1
=====

openroad>
```



The floorplan stage contains physical-only cells that do not perform logic functions.

The identified physical-only cells are:

- sky130_fd_sc_hd__decap_3 (Decoupling Capacitor Cell)
- sky130_fd_sc_hd__tapvpwrvgnd_1 (Tap Cell)

Decap cells improve power supply stability by reducing voltage fluctuations, while Tap cells provide substrate and well connections to VDD/VSS, helping prevent latch-up.

b. Placement

i. Highlight the buffers and inverters in the design

The script used to identify buffers and inverter is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placeme
nt/up_down_counter.def
puts ""
puts "===== BUFFERS ====="
set buffer_list {}
foreach inst [[ord::get_db_block] getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {buf} $cell]} {
        if {[lsearch -exact $buffer_list $cell] == -1} {
            lappend buffer_list $cell
            puts $cell
        }
    }
}
puts ""
puts "===== INVERTERS ====="

set inverter_list {}
foreach inst [[ord::get_db_block] getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {inv} $cell]} {
        if {[lsearch -exact $inverter_list $cell] == -1} {
            lappend inverter_list $cell
            puts $cell
        }
    }
}
}
```



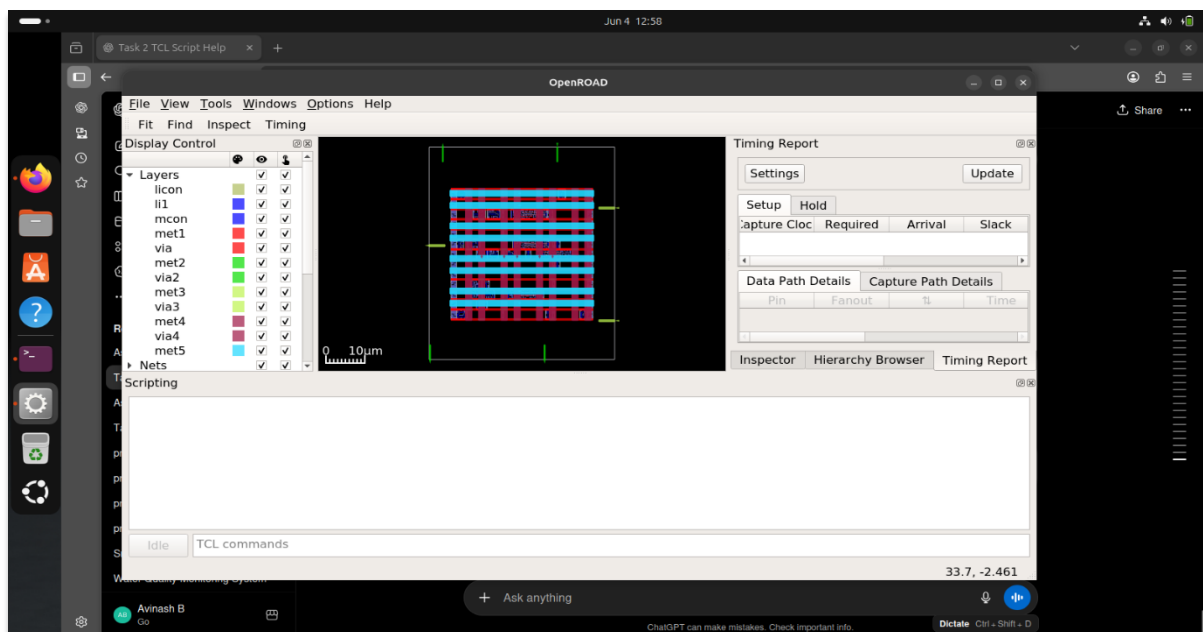
Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

```
puts ""
puts "===== "
gui::show
```

```
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source buffers.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 67 components and 311 component-terminals.
[INFO ODB-0132] Created 2 special nets and 238 connections.
[INFO ODB-0133] Created 29 nets and 73 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def

===== BUFFERS =====
sky130_fd_sc_hd_clkbuf_2
sky130_fd_sc_hd_buf_2
sky130_fd_sc_hd_clkbuf_4

===== INVERTERS =====
sky130_fd_sc_hd_inv_2
```



The script to get instances is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placeme
nt/up_down_counter.def
```



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```
puts "\n===== BUFFER/INVERTER INSTANCES ====="
foreach inst [[ord::get_db_block] getInsts] {
    set inst_name [$inst getName]
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {buf|inv} $cell]} {
        puts "$inst_name --> $cell"
    }
}
```

```
OpenLane Container (ff5509f):/openlane$ vi instances.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '///.tclsh-history' is not writable.
openroad> source instances.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 67 components and 311 component-terminals.
[INFO ODB-0132] Created 2 special nets and 238 connections.
[INFO ODB-0133] Created 29 nets and 73 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def

===== BUFFER/INVERTER INSTANCES =====
_16_ --> sky130_fd_sc_hd__inv_2
_28_ --> sky130_fd_sc_hd__inv_2
_29_ --> sky130_fd_sc_hd__inv_2
_30_ --> sky130_fd_sc_hd__inv_2
_31_ --> sky130_fd_sc_hd__inv_2
input1 --> sky130_fd_sc_hd__clkbuf_2
input2 --> sky130_fd_sc_hd__clkbuf_2
output3 --> sky130_fd_sc_hd__buf_2
output4 --> sky130_fd_sc_hd__buf_2
output5 --> sky130_fd_sc_hd__clkbuf_4
output6 --> sky130_fd_sc_hd__clkbuf_4
openroad>
```

Instance	Cell Type
16	sky130_fd_sc_hd__inv_2
28	sky130_fd_sc_hd__inv_2
29	sky130_fd_sc_hd__inv_2
30	sky130_fd_sc_hd__inv_2
31	sky130_fd_sc_hd__inv_2



Instance	Cell Type
input1	sky130_fd_sc_hd__clkbuf_2
input2	sky130_fd_sc_hd__clkbuf_2
output3	sky130_fd_sc_hd__buf_2
output4	sky130_fd_sc_hd__buf_2
output5	sky130_fd_sc_hd__clkbuf_4
output6	sky130_fd_sc_hd__clkbuf_4

ii. Highlight the flops in the design

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
puts "\n==== FLIP-FLOP INSTANCES ====="
foreach inst [[ord::get_db_block] getInsts] {
    set inst_name [$inst getName]
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {df} $cell]} {
        puts "$inst_name --> $cell"
    }
}
```

```
OpenLane Container (ff5509f):/openlane$ vi ff.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source ff.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 67 components and 311 component-terminals.
[INFO ODB-0132] Created 2 special nets and 238 connections.
[INFO ODB-0133] Created 29 nets and 73 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def

==== FLIP-FLOP INSTANCES =====
_32_ --> sky130_fd_sc_hd__dfrtp_1
_33_ --> sky130_fd_sc_hd__dfrtp_1
_34_ --> sky130_fd_sc_hd__dfrtp_1
_35_ --> sky130_fd_sc_hd__dfrtp_1
openroad> |
```



iii. Capture the site row and measure the height and width

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placeme
nt/up_down_counter.def
set block [ord::get_db_block]
puts "\n===== SITE ROW DETAILS ====="
foreach row [$block getRows] {
    set site [$row getSite]
    puts "Site Name : [$site getName]"
    puts "Site Width : [$site getWidth]"
    puts "Site Height : [$site getHeight]"
    break
}
puts "=====
```

```
OpenLane Container (ff5509f):/openlane$ vi height.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source height.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 67 components and 311 component-terminals.
[INFO ODB-0132] Created 2 special nets and 238 connections.
[INFO ODB-0133] Created 29 nets and 73 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def

===== SITE ROW DETAILS =====
Site Name : unithd
Site Width : 460
Site Height : 2720
=====
```

Parameter	Value
Site Name	unithd
Site Width	460 DBU
Site Height	2720 DBU



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The placement rows are constructed using the unithd site definition from the SKY130 HD standard-cell library. The measured site width is 460 DBU and the site height is 2720 DBU. Standard cells are placed by aligning them to these site dimensions.

iv. Find the different library cells height and width (Ex : AND, OR, FF etc)

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placeme
nt/up_down_counter.def
puts "\n===== LIBRARY CELL DIMENSIONS ====="
set_printed_cells {}
foreach inst [[ord::get_db_block] getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {and|or|df} $cell]} {
        if {[lsearch -exact $printed_cells $cell] == -1} {
            lappend printed_cells $cell
            puts ""
            puts "Cell Name : $cell"
            puts "Width    : [$master getWidth]"
            puts "Height   : [$master getHeight]"
        }
    }
}
puts "\n=====
```

Library Cell	Width (DBU)	Height (DBU)
sky130_fd_sc_hd__xor2_1	3220	2720
sky130_fd_sc_hd__xnor2_1	3220	2720
sky130_fd_sc_hd__or2b_1	2760	2720
sky130_fd_sc_hd__and2b_1	2760	2720
sky130_fd_sc_hd__dfrtp_1	9200	2720

Different standard cells used in the design were analyzed during the placement stage. The AND, OR, XOR, XNOR, and Flip-Flop cells were measured. All cells have a common height



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of 2720 DBU, matching the placement site height, while their widths vary depending on the complexity of the logic function. The flip-flop cell (sky130_fd_sc_hd__dftrp_1) occupies the largest width of 9200 DBU.

```
OpenLane Container (ff5509f):/openlane$ vi library.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source library.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 67 components and 311 component-terminals.
[INFO ODB-0132] Created 2 special nets and 238 connections.
[INFO ODB-0133] Created 29 nets and 73 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/placement/up_down_counter.def

===== LIBRARY CELL DIMENSIONS =====

Cell Name : sky130_fd_sc_hd_xor2_1
Width      : 3220
Height     : 2720

Cell Name : sky130_fd_sc_hd_xnor2_1
Width      : 3220
Height     : 2720

Cell Name : sky130_fd_sc_hd_or2b_1
Width      : 2760
Height     : 2720

Cell Name : sky130_fd_sc_hd_and2b_1
Width      : 2760
Height     : 2720

Cell Name : sky130_fd_sc_hd_dftrp_1
Width      : 9200
Height     : 2720

=====
openroad>
```



c. CTS

i. Highlight the CTS added buffer/inverter

The script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def
puts "\n===== CTS BUFFER / INVERTER INSTANCES ====="
foreach inst [[ord::get_db_block] getInsts] {
    set inst_name [$inst getName]
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {buf|inv} $cell]} {
        puts "$inst_name --> $cell"
    }
}
puts "\n=====
```

```
OpenLane Container (ff5509f):/openlane$ vi cts_buffers.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source cts_buffers.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 70 components and 329 component-terminals.
[INFO ODB-0132] Created 2 special nets and 250 connections.
[INFO ODB-0133] Created 32 nets and 79 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def

===== CTS BUFFER / INVERTER INSTANCES =====
_16_ --> sky130_fd_sc_hd_inv_2
_28_ --> sky130_fd_sc_hd_inv_2
_29_ --> sky130_fd_sc_hd_inv_2
_30_ --> sky130_fd_sc_hd_inv_2
_31_ --> sky130_fd_sc_hd_inv_2
clkbuf_0_clk --> sky130_fd_sc_hd_clkbuf_16
clkbuf_1_0_f_clk --> sky130_fd_sc_hd_clkbuf_16
clkbuf_1_1_f_clk --> sky130_fd_sc_hd_clkbuf_16
input1 --> sky130_fd_sc_hd_clkbuf_2
input2 --> sky130_fd_sc_hd_clkbuf_2
output3 --> sky130_fd_sc_hd_buf_2
output4 --> sky130_fd_sc_hd_buf_2
output5 --> sky130_fd_sc_hd_clkbuf_4
output6 --> sky130_fd_sc_hd_clkbuf_4

=====
openroad> |
```



Instance Name	Cell Type
clkbuf_0_clk	sky130_fd_sc_hd__clkbuf_16
clkbuf_1_0__f_clk	sky130_fd_sc_hd__clkbuf_16
clkbuf_1_1__f_clk	sky130_fd_sc_hd__clkbuf_16

Comparison between the Placement and CTS stages shows that Clock Tree Synthesis inserted three clock buffers of type sky130_fd_sc_hd__clkbuf_16 to distribute the clock signal and reduce clock skew. No additional inverter cells were inserted during CTS.

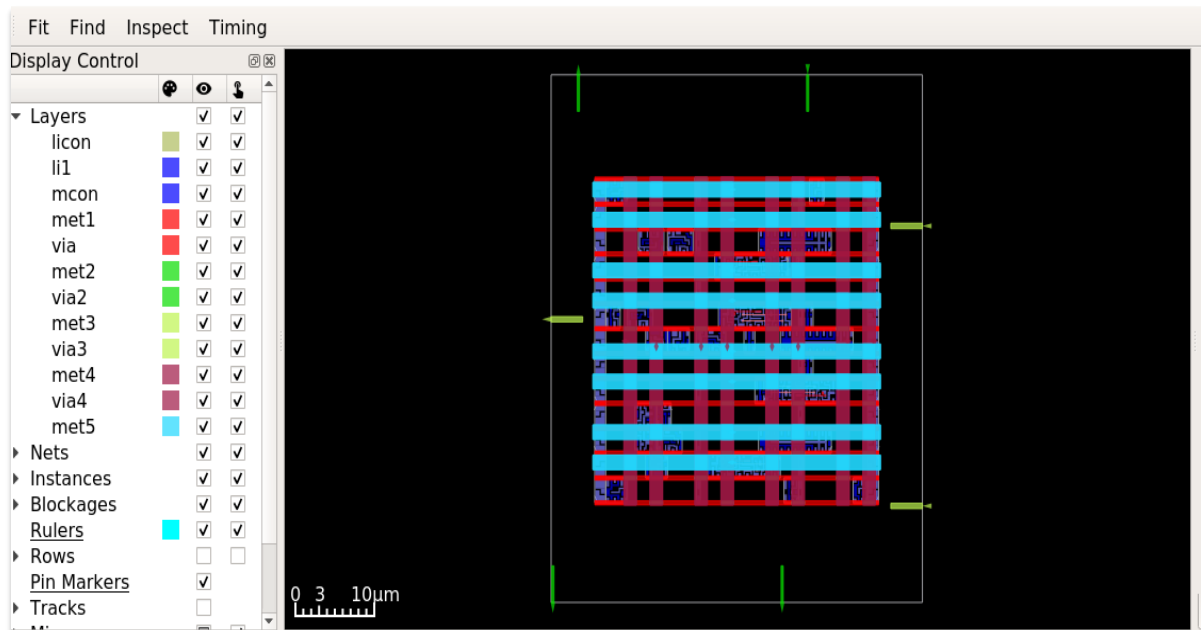
ii. Highlight the Clock routing and its metal layer usage

The script to identify the clock nets:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_d
own_counter.def
puts "\n==== CLOCK RELATED NETS ====="
foreach net [[ord::get_db_block] getNets] {
    set net_name [$net getName]
    if {[regexp -nocase {clk|clock} $net_name]} {
        puts $net_name
    }
}
puts "\n===== "
gui::show
```

```
OpenLane Container (ff5509f):/openlane$ vi routing.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '///.tclsh-history' is not writable.
openroad> source routing.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 70 components and 329 component-terminals.
[INFO ODB-0132] Created 2 special nets and 250 connections.
[INFO ODB-0133] Created 32 nets and 79 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def

==== CLOCK RELATED NETS =====
clk
clknet_0_clk
clknet_1_0__leaf_clk
clknet_1_1__leaf_clk
```



The TCL script to extract the routing layers used by the clock nets is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef

read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_c
ounter.def

puts "\n===== CLOCK NETS ====="

foreach net [[ord::get_db_block] getNets] {
    set net_name [$net getName]
    if {[regexp -nocase {clk|clock} $net_name]} {
        puts $net_name
    }
}

puts "\n===== CTS CLOCK BUFFERS ====="

foreach inst [[ord::get_db_block] getInsts] {
    set inst_name [$inst getName]
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {clkbuf} $cell]} {
```



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```
puts "$inst_name --> $cell"
```

```
}
```

```
}
```

```
OpenLane Container (ff5509f):/openlane$ vi clock.tcl
OpenLane Container (ff5509f):/openlane$ vi clock_layers.tcl
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source clock_layers.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 70 components and 329 component-terminals.
[INFO ODB-0132] Created 2 special nets and 250 connections.
[INFO ODB-0133] Created 32 nets and 79 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/cts/up_down_counter.def

==== CLOCK NETS ====
clk
clknet_0_clk
clknet_1_0__leaf_clk
clknet_1_1__leaf_clk

==== CTS CLOCK BUFFERS ====
clkbuf_0_clk --> sky130_fd_sc_hd__clkbuf_16
clkbuf_1_0__f_clk --> sky130_fd_sc_hd__clkbuf_16
clkbuf_1_1__f_clk --> sky130_fd_sc_hd__clkbuf_16
input1 --> sky130_fd_sc_hd__clkbuf_2
input2 --> sky130_fd_sc_hd__clkbuf_2
output5 --> sky130_fd_sc_hd__clkbuf_4
output6 --> sky130_fd_sc_hd__clkbuf_4
openroad>
```

Clock routing analysis after CTS identified the clock network consisting of clk, clknet_0_clk, clknet_1_0__leaf_clk, and clknet_1_1__leaf_clk. Clock Tree Synthesis inserted three sky130_fd_sc_hd__clkbuf_16 buffers (clkbuf_0_clk, clkbuf_1_0__f_clk, and clkbuf_1_1__f_clk) to distribute the clock signal to the flip-flops and reduce clock skew. Metal layer usage was observed from the CTS GUI routing view.

d. Routing

i. Highlight the signal routing and its metal layers

The TCL script used is:

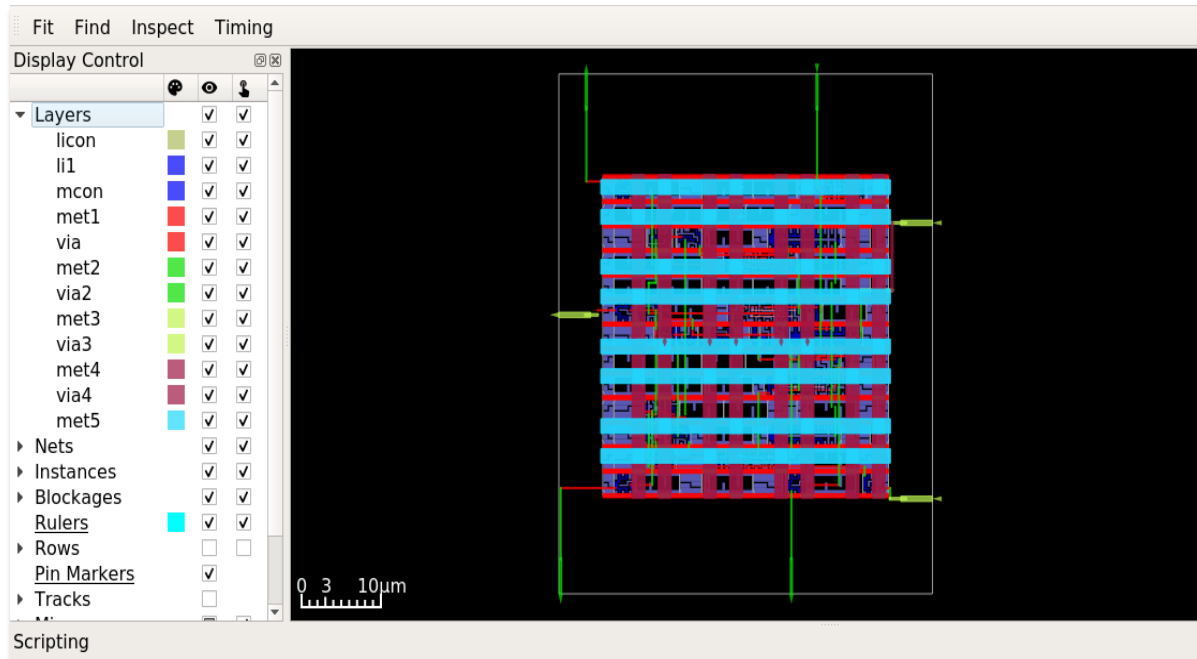
```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef
```



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```
read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/routing/
up_down_counter.def
puts "\n===== SIGNAL NETS ====="
foreach net [[ord::get_db_block] getNets] {
    set net_name [$net getName]
    if {[regexp -nocase {clk|clock} $net_name]} {
        puts $net_name
    }
}
puts "\n===== "
gui::show
```

```
===== SIGNAL NETS =====
VCND
VPWR
count[0]
count[1]
count[2]
count[3]
rst
up_down
_00_
_01_
_02_
_03_
_04_
_05_
_06_
_07_
_08_
_09_
_10_
_11_
_12_
_13_
_14_
_15_
net1
net10
net2
net3
net4
net5
net6
net7
net9
=====
```



The routed design contains signal nets corresponding to the 4-bit counter outputs (count[0] to count[3]), control signals (rst, up_down), and internally generated nets. Signal routing was completed after CTS and is visible in the routing-stage GUI.

ii. Figure out the newly added buffer/inverter in routing stage

The TCL script used is:

```
read_lef
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.no
m.lef

read_def
/openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/routing/
up_down_counter.def

puts "\n===== ROUTING BUFFER / INVERTER INSTANCES ====="

foreach inst [[ord::get_db_block] getInsts] {
    set inst_name [$inst getName]
    set master [$inst getMaster]
    set cell [$master getName]
    if {[regexp -nocase {buf|inv} $cell]} {
        puts "$inst_name --> $cell"
    }
}
```



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```
}
}

puts "\n=====
```

```
openroad> source routing_buffers.tcl
[INFO ODB-0222] Reading LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/routing/up_down_counter.def
[INFO ODB-0128] Design: up_down_counter
[INFO ODB-0130] Created 9 pins.
[INFO ODB-0131] Created 163 components and 707 component-terminals.
[INFO ODB-0132] Created 2 special nets and 622 connections.
[INFO ODB-0133] Created 35 nets and 85 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/up_down_counter/runs/RUN_2026.05.28_02.09.37/results/routing/up_down_counter.def

===== ROUTING BUFFER / INVERTER INSTANCES =====
_16 --> sky130_fd_sc_hd_inv_2
_28 --> sky130_fd_sc_hd_inv_2
_29 --> sky130_fd_sc_hd_inv_2
_30 --> sky130_fd_sc_hd_inv_2
_31 --> sky130_fd_sc_hd_inv_2
clkbuf_0_clk --> sky130_fd_sc_hd_clkbuf_16
clkbuf_1_0_f_clk --> sky130_fd_sc_hd_clkbuf_16
clkbuf_1_1_f_clk --> sky130_fd_sc_hd_clkbuf_16
input1 --> sky130_fd_sc_hd_clkbuf_2
input2 --> sky130_fd_sc_hd_clkbuf_2
output3 --> sky130_fd_sc_hd_clkbuf_4
output4 --> sky130_fd_sc_hd_clkbuf_4
output5 --> sky130_fd_sc_hd_clkbuf_4
output6 --> sky130_fd_sc_hd_clkbuf_4
```

Category	Observation
New Buffers Added During Routing	None
New Inverters Added During Routing	None
CTS Buffers Retained	clkbuf_0_clk, clkbuf_1_0__f_clk, clkbuf_1_1__f_clk
Existing Inverters Retained	16, 28, 29, 30, 31

Comparison of CTS and Routing stage databases shows that no additional buffers or inverters were inserted during routing. The clock buffers inserted during CTS (clkbuf_0_clk, clkbuf_1_0__f_clk, and clkbuf_1_1__f_clk) were retained, and the inverter instances remained unchanged. Therefore, routing completed without requiring additional buffer or inverter insertion.



3. Setup the flow with below features and capture the proof for each from the design/log/reports.

a. do PnR for 32 bit up/down counter 6ns clock period

PnR of 32-bit up/down counter with 6 ns clock, the TCL script that successfully runs the complete OpenLane flow is:

```
package require openlane 1.0.2
prep -design up_down_counter
run_synthesis
run_floorplan
run_placement
run_cts
run_routing
run_magic
run_magic_spice_export
run_parasitics_sta
```

To do PnR we need to edit config.json is updated to:

```
{
  "DESIGN_NAME": "up_down_counter",
  "VERILOG_FILES": "dir::src/up_down_counter.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "FP_CORE_UTIL": 20,
  "DIE_AREA": "0 0 100 100"
}
```

```
[STEP 32]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_04.11.48/logs/signoff/32-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_04.11.48/tmp/16-up_down_counter.sdc
% 
```

b. Set the setup uncertainty of the design to 10 % of your clock frequency

The modified script used is:

```
package require openlane 1.0.2
prep -design up_down_counter
set ::env(SYNTH_CLOCK_UNCERTAINTY) 0.6
run_synthesis
run_floorplan
run_placement
run_cts
```



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```
run_routing
run_magic
run_magic_spice_export
run_parasitics_sta
```

```
[STEP 30]
[INFO]: Running SPEF Extraction at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_04.42.56/logs/signoff/30-parasitics_extraction.nom.log)...
[STEP 31]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_04.42.56/logs/signoff/31-rcx_mcsta.nom.log)...
[STEP 32]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_04.42.56/logs/signoff/32-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_04.42.56/tmp/16-up_down_counter.sdc
% exit
OpenLane Container (ff5509f):/openlane$ cat /openlane/designs/up_down_counter/runs/RUN_2026.06.06_04.42.56/tmp/16-up_down_counter.sdc | grep uncertainty
set_clock_uncertainty 0.6000 clk
OpenLane Container (ff5509f):/openlane$
```

c. Set derate value to 3%

The updated TCL Script is:

```
package require openlane 1.0.2
prep -design up_down_counter
set ::env(SYNTH_CLOCK_UNCERTAINTY) 0.6
set ::env(SYNTH_TIMING_DERATE) 0.03
run_synthesis
run_floorplan
run_placement
run_cts
run_routing
run_magic
run_magic_spice_export
run_parasitics_sta
```

```
[STEP 32]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_04.50.58/logs/signoff/32-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_04.50.58/tmp/16-up_down_counter.sdc
OpenLane Container (ff5509f):/openlane$ cat /openlane/designs/up_down_counter/runs/RUN_2026.06.06_04.50.58/tmp/16-up_down_counter.sdc | grep derate
set_timing_derate -early 0.9700
set_timing_derate -late 1.0300
OpenLane Container (ff5509f):/openlane$
```

d. Utilization of floorplan to 70%

The updated TCL script is:

```
package require openlane 1.0.2
prep -design up_down_counter
set ::env(SYNTH_CLOCK_UNCERTAINTY) 0.6
set ::env(SYNTH_TIMING_DERATE) 0.03
set ::env(FP_CORE_UTIL) 70
set ::env(PL_TARGET_DENSITY) 0.77
run_synthesis
```



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run_floorplan
run_placement
run_cts
run_routing
run_magic
run_magic_spice_export
run_parasitics_sta

```
[STEP 32]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/logs/signoff/32-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd_tapvpmrvngnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/tmp/16-up_down_counter.sdc
% cat /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/logs/placement/7-global.log
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
[INFO]: Reading ODB at '/openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/results/floorplan/up_down_counter.odb'...
define_corners Typical
read_liberty -corner Typical /home/lucifero/.ciel/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd_tt_025C_1v80.lib
Using 1e-12 for capacitance...
Using 1e+03 for resistance...
Using 1e-09 for time...
Using 1e+00 for voltage...
Using 1e-03 for current...
Using 1e-09 for power...
Using 1e-06 for distance...
Reading design constraints file at '/openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/tmp/floorplan/3-initial_fp.sdc'...
[INFO]: Setting RC values...
[INFO]: Setting signal min routing layer to: met1 and clock min routing layer to met3.
[INFO]: Setting signal max routing layer to: met5 and clock max routing layer to met5.
[INFO GPL-0002] DBU: 1000
[INFO GPL-0003] SiteSize: 460 2720
[INFO GPL-0004] CoreAreaLxLy: 5520 10880
[INFO GPL-0005] CoreAreaUxUy: 67160 70720
[INFO GPL-0006] NumInstances: 321
[INFO GPL-0007] NumPlaceInstances: 229
[INFO GPL-0008] NumFixedInstances: 92
[INFO GPL-0009] NumDummyInstances: 0
[INFO GPL-0010] NumNets: 232
[INFO GPL-0011] NumPins: 777
[INFO GPL-0012] DieAreaLxLy: 0 0
[INFO GPL-0013] DieAreaUxUy: 72740 83460
[INFO GPL-0014] CoreAreaLxLy: 5520 10880
[INFO GPL-0015] CoreAreaUxUy: 67160 70720
[INFO GPL-0016] CoreArea: 3688537600
[INFO GPL-0017] NonPlaceInstsArea: 225216000
[INFO GPL-0018] PlaceInstsArea: 2665056000
[INFO GPL-0019] Util(%): 76.95
[INFO GPL-0020] StdInstsArea: 2665056000
[INFO GPL-0021] MacroInstsArea: 0
[InitialPlace] Iter: 1 CC residual: 0.00000010 HPWL: 4626060
```

```
OpenLane Container (ff5509f):/openlane$ grep "Util" /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/logs/placement/7-global.log
[INFO GPL-0019] Util(%): 76.95
OpenLane Container (ff5509f):/openlane$
```

e. Insert two column of tap cells

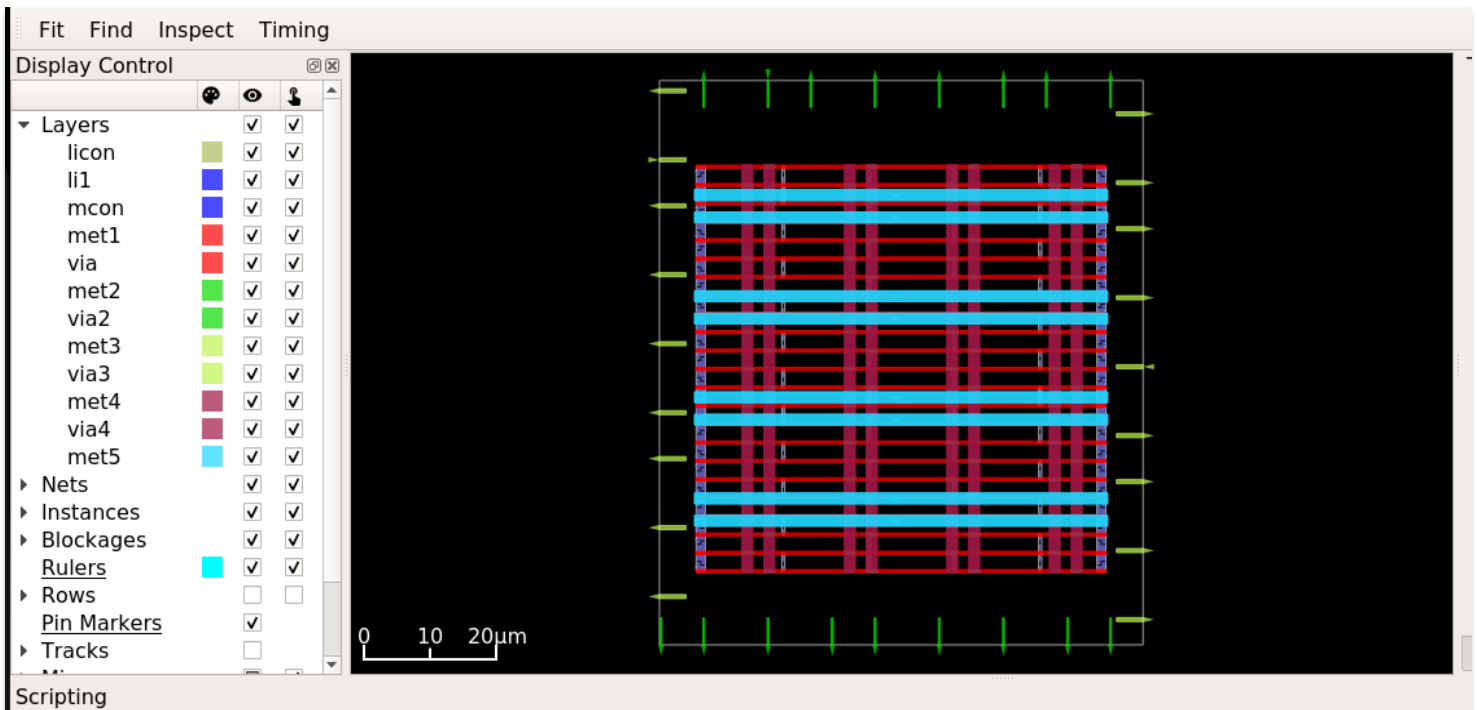
Tap-cell insertion was performed during the floor planning stage using the OpenLane automatic tap-cell insertion flow. The generated log reported:



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[INFO TAP-0005] Inserted 48 tapcells.

This confirms successful insertion of tap cells throughout the design. Multiple tap-cell columns were automatically generated across the core area according to the configured tap-cell spacing rules

```
OpenLane Container (ff5509f):/openlane$ grep FP_TAPCELL_DIST /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/config.tcl
set ::env(FP_TAPCELL_DIST) "13"
OpenLane Container (ff5509f):/openlane$ grep -i "tapcell\\|tap" /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/logs/floorplan/5-tap.log | tail -30
[INFO TAP-0004] Inserted 44 endcaps.
[INFO TAP-0005] Inserted 48 tapcells.
Writing OpenROAD database to '/openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/tmp/floorplan/5-tapcell.odb'...
Writing layout to '/openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/tmp/floorplan/5-tapcell.def'...
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> read_db /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/results/floorplan/up_down_counter.odb
openroad> gui::show
[WARNING GUI-0076] QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-'
libGL error: unable to load driver: swrast_dri.so
libGL error: failed to load driver: swrast
OpenLane Container (ff5509f):/openlane$
```





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f. Enable core ring in the design

We must add FP_PDN_CORE_RING entry in config.json

```
{
  "DESIGN_NAME": "up_down_counter",
  "VERILOG_FILES": "dir::src/up_down_counter.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "FP_CORE_UTIL": 20,
  "FP_PDN_CORE_RING": true,
  "DIE_AREA": "0 0 100 100"
}
```

```
penLane Container (ff5509f):/openlane$ grep FP_PDN_CORE_RING /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.39.52/config.tcl
et ::env(FP_PDN_CORE_RING) "0"
et ::env(FP_PDN_CORE_RING_HOFFSET) "6"
et ::env(FP_PDN_CORE_RING_HSPACING) "1.7"
et ::env(FP_PDN_CORE_RING_HWIDTH) "1.6"
et ::env(FP_PDN_CORE_RING_VOFFSET) "6"
et ::env(FP_PDN_CORE_RING_VSPACING) "1.7"
et ::env(FP_PDN_CORE_RING_VWIDTH) "1.6"
penLane Container (ff5509f):/openlane$ vi openroad.tcl
penLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
penLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
ll rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
available under the Apache License, version 2.0. See the LICENSE file for more details.
```

```
source openroad.tcl
INFO]: Using configuration in 'designs/up_down_counter/config.json'...
INFO]: PDK Root: /home/lucifero/.ciel
INFO]: Process Design Kit: sky130A
INFO]: Standard Cell Library: sky130_fd_sc_hd
INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
INFO]: Run Directory: /openlane/designs/up_down_counter/runs/RUN_2026.06.06_10.59.43
INFO]: Saving runtime environment...
INFO]: Preparing LEF files for the nom corner...
INFO]: Preparing LEF files for the min corner...
INFO]: Preparing LEF files for the max corner...
```

```
[STEP 29]
[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/signoff/29-rcx_mcsta.max.log)...
[STEP 30]
[INFO]: Running SPEF Extraction at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/signoff/30-parasitics_extraction.nom.log)...
[STEP 31]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/signoff/31-rcx_mcsta.nom.log)...
[STEP 32]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/signoff/32-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd_tapvpwrvngnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd_fill_2 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd_decap_12 blackboxed during sta
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/16-up_down_counter.sdc
% grep FP_PDN_CORE_RING /openlane/designs/up_down_counter/runs/<actual_new_run>/config.tcl
/usr/bin/grep: /openlane/designs/up_down_counter/runs/<actual_new_run>/config.tcl: No such file or directory
child process exited abnormally
% exit
OpenLane Container (ff5509f):/openlane$ grep FP_PDN_CORE_RING /openlane/designs/up_down_counter/runs/<actual_new_run>/config.tcl
bash: actual_new_run: No such file or directory
OpenLane Container (ff5509f):/openlane$ grep FP_PDN_CORE_RING /openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/config.tcl
set ::env(FP_PDN_CORE_RING) "1"
set ::env(FP_PDN_CORE_RING_HOFFSET) "6"
set ::env(FP_PDN_CORE_RING_HSPACING) "1.7"
set ::env(FP_PDN_CORE_RING_HWIDTH) "1.6"
set ::env(FP_PDN_CORE_RING_VOFFSET) "6"
set ::env(FP_PDN_CORE_RING_VSPACING) "1.7"
set ::env(FP_PDN_CORE_RING_VWIDTH) "1.6"
OpenLane Container (ff5509f):/openlane$
```



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Core ring generation was enabled by setting FP_PDN_CORE_RING to 1. The generated configuration confirms the core ring was enabled and the PDN was generated with a core power ring.

g. CTS target skew to 5% of the clock period.

The design clock period is 6 ns. The assignment requires a clock skew target equal to 5% of the clock period.

Target skew calculation:

$$5\% \times 6 \text{ ns} = 0.3 \text{ ns}$$

The OpenLane v1.0.2 release used in this work does not support the CTS_TARGET_SKEW configuration variable, as the parameter has been removed from the tool flow. Therefore, Clock Tree Synthesis (CTS) was executed using the default CTS flow and the achieved clock skew was extracted from the CTS timing report.

```
OpenLane Container (ff5509f):/openlane$ find /openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30 -name "*cts*" -o -name "*clock*"
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/results/cts
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts.lib.exclude.list
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts.lib
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts-slowest.lib
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts-fastest.lib
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts-fastest.lib.exclude.list
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/tmp/cts/cts-slowest.lib.exclude.list
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/reports/cts
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/12-cts.log
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/12-cts.warnings
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/12-cts.errors
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/13-cts_sta.log
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/13-cts_sta.errors
/openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/13-cts_sta.warnings
OpenLane Container (ff5509f):/openlane$ grep -i "skew" /openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/12-cts.log
OpenLane Container (ff5509f):/openlane$ grep -i "skew" /openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/13-cts_sta.log
skew_report
report_clock_skew
Latency      CRPR      Skew
skew_report_end
OpenLane Container (ff5509f):/openlane$ sed -n '/skew_report/,/skew_report_end/p' /openlane/designs/up_down_counter/runs/RUN_2026.06.06_11.23.30/logs/cts/13-cts_sta.log
skew_report
=====
report_clock_skew
=====
Clock clk
Latency      CRPR      Skew
_414_/CLK ^
0.34
_424_/CLK ^
0.31      -0.01      0.01
skew_report_end
OpenLane Container (ff5509f):/openlane$
```



ASSIGNMENT 3

Take Riscv 32 bit RTL

1. Uncertainty at synthesis 15% of clock period and from CTS 5% of clk period

The config.json file used is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The script used to set the uncertainty is:

```
set sdc_file "$::env(RUN_DIR)/results/cts/picorv32.sdc"

set fp [open $sdc_file r]
set data [read $fp]
close $fp

regsub -all {set_clock_uncertainty 0\.9000 clk} $data {set_clock_uncertainty 0.3000 clk} data

set fp [open $sdc_file w]
puts $fp $data
close $fp
```



After setting the clock uncertainty use the commands:

- `prep -designs riscv`
- `run_synthesis`
- `run_floorplan`
- `run_cts`
- After using `run_cts` check the CTS uncertainty is 5% of clk period if its 5% of clock period then use:
- `run_routing`

Then use the script to get uncertainty at synthesis and CTS:

```
catch {exec grep -R "set_clock_uncertainty" $::env(RUN_DIR)} result1
puts $result1

puts "\n===== TIMING DERATE ====="
catch {exec grep -R "set_timing_derate" $::env(RUN_DIR)} result2
puts $result2
```

The Output of this script is:

```
% vi uncertainty_derate.tcl
% source uncertainty_derate.tcl

===== CLOCK UNCERTAINTY =====
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/results/cts/picorv32.sdc;set_clock_uncertainty 0.3000 clk
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/placement/9-resizer.sdc;set_clock_uncertainty 0.9000 clk
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/floorplan/3-initial_fp.sdc;set_clock_uncertainty 0.9000 clk

===== TIMING DERATE =====
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/results/cts/picorv32.sdc;set_timing_derate -early 0.9500
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/results/cts/picorv32.sdc;set_timing_derate -late 1.0500
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/placement/9-resizer.sdc;set_timing_derate -early 0.9500
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/placement/9-resizer.sdc;set_timing_derate -late 1.0500
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/floorplan/3-initial_fp.sdc;set_timing_derate -early 0.9500
/openlane/designs/riscv/runs/RUN_2026.06.22_04.41.57/tmp/floorplan/3-initial_fp.sdc;set_timing_derate -late 1.0500
%
```

- Clock Period: 6 ns
- Synthesis Uncertainty: 0.9000 ns
- CTS Uncertainty: 0.3000 ns
- Timing Derate (Early): 0.9500
- Timing Derate (Late): 1.0500



2. Set derate 3% throughout the design

The config.json must be modified to:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

After saving the config.json file use run_synthesis and then execute the command:

```
puts "\n===== DERATE CONFIGURATION ====="
exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
```

You will get the output:

```
% prep -design riscv -tag DERATE3 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/DERATE3
[INFO]: Removing existing /openlane/designs/riscv/runs/DERATE3...
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/DERATE3/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/DERATE3/logs/synthesis/2-sta.log)...
% puts $::env(RUN_DIR)
/openlane/designs/riscv/runs/DERATE3
% exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
set ::env(SYNTH_TIMING_DERATE) "0.03"
% exec grep "SYNTH_TIMING_DERATE" $::env(RUN_DIR)/config.tcl
set ::env(SYNTH_TIMING_DERATE) "0.03"
% exec find $::env(RUN_DIR) -name "*.sdc"
/openlane/designs/riscv/runs/DERATE3/tmp/synthesis/synthesis.sdc
% exec grep "timing_derate" $::env(RUN_DIR)/tmp/synthesis/synthesis.sdc
child process exited abnormally
% exec cat $::env(RUN_DIR)/tmp/synthesis/synthesis.sdc
set_driving_cell sky130_fd_sc_hd__inv_2
set_load 33.442
% vi derate.tcl
% source derate.tcl

===== DERATE CONFIGURATION =====
set ::env(SYNTH_TIMING_DERATE) "0.03"
%
```



3. Set input and output delay 50% of clk time period

The updated config.json file is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The base.sdc file is created and its content is:

```
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.9 [get_clocks clk]

set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]
~
~
```

The use run_synthesis and verify the logs and finally the output is:

```
OpenLane Container (ff5509f):/openlane$ vi designs/riscv/config.json
OpenLane Container (ff5509f):/openlane$ vi /openlane/designs/riscv/base.sdc
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design riscv -tag IO50_SDC -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/IO50_SDC
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/IO50_SDC/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/IO50_SDC/logs/synthesis/2-sta.log)...
```



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```
% exec grep -R "BASE_SDC_FILE" $::env(RUN_DIR)
/openlane/designs/riscv/runs/I050_SDC/config.tcl:set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
/openlane/designs/riscv/runs/I050_SDC/config_in.tcl:set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
% exec cat /openlane/designs/riscv/base.sdc
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.9 [get_clocks clk]

set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]
```

```
% vi logs.tcl
% source logs.tcl

===== CLOCK / IO CONSTRAINTS =====
create_clock -name clk -period 6 [get_ports clk]

set_clock_uncertainty 0.9 [get_clocks clk]

set_input_delay 3.0 -clock clk [all_inputs]
set_output_delay 3.0 -clock clk [all_outputs]

===== DERATE =====
set ::env(SYNTH_TIMING_DERATE) "0.03"

===== BASE SDC =====
set ::env(BASE_SDC_FILE) "/openlane/designs/riscv/base.sdc"
```

From these logs we can say:

- Clock Period: 6 ns
- Clock Uncertainty: 0.9 ns
- Input Delay: 3.0 ns
- Output Delay: 3.0 ns
- Timing Derate: 0.03 (3%)

Save Point:

- Design: PicoRV32 (32-bit RISC-V)
- Clock Period: 6 ns
- Clock Uncertainty: 0.9 ns
- Input Delay: 3.0 ns
- Output Delay: 3.0 ns
- Timing Derate: 3%
- Constraint File: base.sdc



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4. Utilization should be set 60% at floorplan

The updated config.json script is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 1,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The output we get is:

```
OpenLane Container (ff5509f):/openlane$ vi designs/riscv/config.json
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design riscv -tag UTIL60 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/UTIL60
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/UTIL60/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/UTIL60/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv/runs/UTIL60/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 408.94 and height 408.0.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv/runs/UTIL60/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv/runs/UTIL60/logs/floorplan/6-pdn.log)...
% puts [exec grep "FP_CORE_UTIL" $::env(RUN_DIR)/config.tcl]
set ::env(FP_CORE_UTIL) "60"
%
```



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Save Point:

- Design: PicoRV32 (32-bit RISC-V)
- Floorplan Utilization: 60%
- Evidence File: runs/UTIL60/config.tcl

5. Floorplan ratio should be 0.7

The updated config.json:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.35,
  "FP_ASPECT_RATIO": 0.7,
  "FP_IO_MODE": 1,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

The TCL script used is:

```
puts "\n==== FLOORPLAN ASPECT RATIO ====="
puts [exec grep "FP_ASPECT_RATIO" $::env(RUN_DIR)/config.tcl]
```



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The output is:

```
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design riscv -tag AR07 -overwrite
[INFO]: Using configuration in 'designs/riscv/config.json'...
[INFO]: PDK Root: /home/lucifero/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv/runs/AR07
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv/runs/AR07/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv/runs/AR07/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv/runs/AR07/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 488.52 and height 340.0.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv/runs/AR07/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv/runs/AR07/logs/floorplan/6-pdn.log)...
% vi floorplan.tcl
% source floorplan.tcl

===== FLOORPLAN ASPECT RATIO =====
set ::env(FP_ASPECT_RATIO) "0.7"
%
```

6. port placement - inputs to one side outputs on other side and place it on metal4

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```



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The file used to do manual port placement is pin_order.cfg and its content is:

```
#BUS_SORT
```

```
#N
```

```
@min_distance=0.1
```

```
clk
```

```
resetn
```

```
mem_ready
```

```
pcpi_wr
```

```
pcpi_wait
```

```
pcpi_ready
```

```
irq\[0\]
```

```
irq\[1\]
```

```
irq\[2\]
```

```
irq\[3\]
```

```
irq\[4\]
```

```
irq\[5\]
```

```
irq\[6\]
```

```
irq\[7\]
```

```
irq\[8\]
```

```
irq\[9\]
```

```
irq\[10\]
```

```
irq\[11\]
```

```
irq\[12\]
```

```
irq\[13\]
```

```
irq\[14\]
```

```
irq\[15\]
```

```
irq\[16\]
```

```
irq\[17\]
```

```
irq\[18\]
```

```
irq\[19\]
```

```
irq\[20\]
```



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irq\[21\]

irq\[22\]

irq\[23\]

irq\[24\]

irq\[25\]

irq\[26\]

irq\[27\]

irq\[28\]

irq\[29\]

irq\[30\]

irq\[31\]

mem_rdata\[0\]

mem_rdata\[1\]

mem_rdata\[2\]

mem_rdata\[3\]

mem_rdata\[4\]

mem_rdata\[5\]

mem_rdata\[6\]

mem_rdata\[7\]

mem_rdata\[8\]

mem_rdata\[9\]

mem_rdata\[10\]

mem_rdata\[11\]

mem_rdata\[12\]

mem_rdata\[13\]

mem_rdata\[14\]

mem_rdata\[15\]

mem_rdata\[16\]

mem_rdata\[17\]

mem_rdata\[18\]

mem_rdata\[19\]



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mem_rdata\[20\]

mem_rdata\[21\]

mem_rdata\[22\]

mem_rdata\[23\]

mem_rdata\[24\]

mem_rdata\[25\]

mem_rdata\[26\]

mem_rdata\[27\]

mem_rdata\[28\]

mem_rdata\[29\]

mem_rdata\[30\]

mem_rdata\[31\]

pcpi_rd\[0\]

pcpi_rd\[1\]

pcpi_rd\[2\]

pcpi_rd\[3\]

pcpi_rd\[4\]

pcpi_rd\[5\]

pcpi_rd\[6\]

pcpi_rd\[7\]

pcpi_rd\[8\]

pcpi_rd\[9\]

pcpi_rd\[10\]

pcpi_rd\[11\]

pcpi_rd\[12\]

pcpi_rd\[13\]

pcpi_rd\[14\]

pcpi_rd\[15\]

pcpi_rd\[16\]

pcpi_rd\[17\]

pcpi_rd\[18\]



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pcpi_rd\[19\]

pcpi_rd\[20\]

pcpi_rd\[21\]

pcpi_rd\[22\]

pcpi_rd\[23\]

pcpi_rd\[24\]

pcpi_rd\[25\]

pcpi_rd\[26\]

pcpi_rd\[27\]

pcpi_rd\[28\]

pcpi_rd\[29\]

pcpi_rd\[30\]

pcpi_rd\[31\]

#S

trap

mem_valid

mem_instr

mem_la_read

mem_la_write

pcpi_valid

trace_valid

ei\[0\]

ei\[1\]

ei\[2\]

ei\[3\]

ei\[4\]

ei\[5\]

ei\[6\]

ei\[7\]

ei\[8\]

ei\[9\]



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eoì\10\]

eoì\11\]

eoì\12\]

eoì\13\]

eoì\14\]

eoì\15\]

eoì\16\]

eoì\17\]

eoì\18\]

eoì\19\]

eoì\20\]

eoì\21\]

eoì\22\]

eoì\23\]

eoì\24\]

eoì\25\]

eoì\26\]

eoì\27\]

eoì\28\]

eoì\29\]

eoì\30\]

eoì\31\]

mem_addr\0\]

mem_addr\1\]

mem_addr\2\]

mem_addr\3\]

mem_addr\4\]

mem_addr\5\]

mem_addr\6\]

mem_addr\7\]

mem_addr\8\]



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mem_addr\[9\]

mem_addr\[10\]

mem_addr\[11\]

mem_addr\[12\]

mem_addr\[13\]

mem_addr\[14\]

mem_addr\[15\]

mem_addr\[16\]

mem_addr\[17\]

mem_addr\[18\]

mem_addr\[19\]

mem_addr\[20\]

mem_addr\[21\]

mem_addr\[22\]

mem_addr\[23\]

mem_addr\[24\]

mem_addr\[25\]

mem_addr\[26\]

mem_addr\[27\]

mem_addr\[28\]

mem_addr\[29\]

mem_addr\[30\]

mem_addr\[31\]

mem_wdata\[0\]

mem_wdata\[1\]

mem_wdata\[2\]

mem_wdata\[3\]

mem_wdata\[4\]

mem_wdata\[5\]

mem_wdata\[6\]

mem_wdata\[7\]



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mem_wdata\[8\]

mem_wdata\[9\]

mem_wdata\[10\]

mem_wdata\[11\]

mem_wdata\[12\]

mem_wdata\[13\]

mem_wdata\[14\]

mem_wdata\[15\]

mem_wdata\[16\]

mem_wdata\[17\]

mem_wdata\[18\]

mem_wdata\[19\]

mem_wdata\[20\]

mem_wdata\[21\]

mem_wdata\[22\]

mem_wdata\[23\]

mem_wdata\[24\]

mem_wdata\[25\]

mem_wdata\[26\]

mem_wdata\[27\]

mem_wdata\[28\]

mem_wdata\[29\]

mem_wdata\[30\]

mem_wdata\[31\]

mem_wstrb\[0\]

mem_wstrb\[1\]

mem_wstrb\[2\]

mem_wstrb\[3\]

mem_la_addr\[0\]

mem_la_addr\[1\]

mem_la_addr\[2\]



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mem_la_addr\[3\
mem_la_addr\[4\
mem_la_addr\[5\
mem_la_addr\[6\
mem_la_addr\[7\
mem_la_addr\[8\
mem_la_addr\[9\
mem_la_addr\[10\
mem_la_addr\[11\
mem_la_addr\[12\
mem_la_addr\[13\
mem_la_addr\[14\
mem_la_addr\[15\
mem_la_addr\[16\
mem_la_addr\[17\
mem_la_addr\[18\
mem_la_addr\[19\
mem_la_addr\[20\
mem_la_addr\[21\
mem_la_addr\[22\
mem_la_addr\[23\
mem_la_addr\[24\
mem_la_addr\[25\
mem_la_addr\[26\
mem_la_addr\[27\
mem_la_addr\[28\
mem_la_addr\[29\
mem_la_addr\[30\
mem_la_addr\[31\
mem_la_wdata\[0\
mem_la_wdata\[1\



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mem_la_wdata\[2\]

mem_la_wdata\[3\]

mem_la_wdata\[4\]

mem_la_wdata\[5\]

mem_la_wdata\[6\]

mem_la_wdata\[7\]

mem_la_wdata\[8\]

mem_la_wdata\[9\]

mem_la_wdata\[10\]

mem_la_wdata\[11\]

mem_la_wdata\[12\]

mem_la_wdata\[13\]

mem_la_wdata\[14\]

mem_la_wdata\[15\]

mem_la_wdata\[16\]

mem_la_wdata\[17\]

mem_la_wdata\[18\]

mem_la_wdata\[19\]

mem_la_wdata\[20\]

mem_la_wdata\[21\]

mem_la_wdata\[22\]

mem_la_wdata\[23\]

mem_la_wdata\[24\]

mem_la_wdata\[25\]

mem_la_wdata\[26\]

mem_la_wdata\[27\]

mem_la_wdata\[28\]

mem_la_wdata\[29\]

mem_la_wdata\[30\]

mem_la_wdata\[31\]

mem_la_wstrb\[0\]



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mem_la_wstrb\[1\]

mem_la_wstrb\[2\]

mem_la_wstrb\[3\]

pcpi_insn\[0\]

pcpi_insn\[1\]

pcpi_insn\[2\]

pcpi_insn\[3\]

pcpi_insn\[4\]

pcpi_insn\[5\]

pcpi_insn\[6\]

pcpi_insn\[7\]

pcpi_insn\[8\]

pcpi_insn\[9\]

pcpi_insn\[10\]

pcpi_insn\[11\]

pcpi_insn\[12\]

pcpi_insn\[13\]

pcpi_insn\[14\]

pcpi_insn\[15\]

pcpi_insn\[16\]

pcpi_insn\[17\]

pcpi_insn\[18\]

pcpi_insn\[19\]

pcpi_insn\[20\]

pcpi_insn\[21\]

pcpi_insn\[22\]

pcpi_insn\[23\]

pcpi_insn\[24\]

pcpi_insn\[25\]

pcpi_insn\[26\]

pcpi_insn\[27\]



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pcpi_insn\[28\]

pcpi_insn\[29\]

pcpi_insn\[30\]

pcpi_insn\[31\]

pcpi_rs1\[0\]

pcpi_rs1\[1\]

pcpi_rs1\[2\]

pcpi_rs1\[3\]

pcpi_rs1\[4\]

pcpi_rs1\[5\]

pcpi_rs1\[6\]

pcpi_rs1\[7\]

pcpi_rs1\[8\]

pcpi_rs1\[9\]

pcpi_rs1\[10\]

pcpi_rs1\[11\]

pcpi_rs1\[12\]

pcpi_rs1\[13\]

pcpi_rs1\[14\]

pcpi_rs1\[15\]

pcpi_rs1\[16\]

pcpi_rs1\[17\]

pcpi_rs1\[18\]

pcpi_rs1\[19\]

pcpi_rs1\[20\]

pcpi_rs1\[21\]

pcpi_rs1\[22\]

pcpi_rs1\[23\]

pcpi_rs1\[24\]

pcpi_rs1\[25\]

pcpi_rs1\[26\]



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pcpi_rs1\[27\]

pcpi_rs1\[28\]

pcpi_rs1\[29\]

pcpi_rs1\[30\]

pcpi_rs1\[31\]

pcpi_rs2\[0\]

pcpi_rs2\[1\]

pcpi_rs2\[2\]

pcpi_rs2\[3\]

pcpi_rs2\[4\]

pcpi_rs2\[5\]

pcpi_rs2\[6\]

pcpi_rs2\[7\]

pcpi_rs2\[8\]

pcpi_rs2\[9\]

pcpi_rs2\[10\]

pcpi_rs2\[11\]

pcpi_rs2\[12\]

pcpi_rs2\[13\]

pcpi_rs2\[14\]

pcpi_rs2\[15\]

pcpi_rs2\[16\]

pcpi_rs2\[17\]

pcpi_rs2\[18\]

pcpi_rs2\[19\]

pcpi_rs2\[20\]

pcpi_rs2\[21\]

pcpi_rs2\[22\]

pcpi_rs2\[23\]

pcpi_rs2\[24\]

pcpi_rs2\[25\]



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pcpi_rs2\[26\]

pcpi_rs2\[27\]

pcpi_rs2\[28\]

pcpi_rs2\[29\]

pcpi_rs2\[30\]

pcpi_rs2\[31\]

trace_data\[0\]

trace_data\[1\]

trace_data\[2\]

trace_data\[3\]

trace_data\[4\]

trace_data\[5\]

trace_data\[6\]

trace_data\[7\]

trace_data\[8\]

trace_data\[9\]

trace_data\[10\]

trace_data\[11\]

trace_data\[12\]

trace_data\[13\]

trace_data\[14\]

trace_data\[15\]

trace_data\[16\]

trace_data\[17\]

trace_data\[18\]

trace_data\[19\]

trace_data\[20\]

trace_data\[21\]

trace_data\[22\]

trace_data\[23\]

trace_data\[24\]



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trace_data\[25\]

trace_data\[26\]

trace_data\[27\]

trace_data\[28\]

trace_data\[29\]

trace_data\[30\]

trace_data\[31\]

trace_data\[32\]

trace_data\[33\]

trace_data\[34\]

trace_data\[35\]

#E

#W

After saving pin_order.cfg use run_synthesis, run_floorplan and run_placement then use this script:

```
#!/usr/bin/env tclsh
```

```
set def_file "/openlane/designs/riscv/runs/GUI_IO/tmp/floorplan/4-io.def"
```

```
set fp [open $def_file r]
```

```
set pin ""
```

```
set direction ""
```

```
set layer ""
```

```
set x ""
```

```
set y ""
```

```
set orient ""
```

```
set inputs 0
```

```
set outputs 0
```

```
set north 0
```

```
set south 0
```

```
set metal4 0
```



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```
puts "=====
puts "      PORT PLACEMENT REPORT"
puts "=====

while {[gets $fp line] >= 0} {
    # New pin definition
    if {[regexp {\^s*\s+(\S+).*DIRECTION\s+(INPUT|OUTPUT)} $line -> pin direction]} {
        if {$direction eq "INPUT"} {
            incr inputs
        } else {
            incr outputs
        }
        continue
    }
    # Metal layer
    if {[regexp {\+ LAYER\s+(\S+)} $line -> layer]} {
        if {$layer eq "met4"} {
            incr metal4
        }
        continue
    }
    # Pin location
    if {[regexp {\+ PLACED\s+\(\s*([0-9]+)\s+([0-9]+)\s*\)\s+([A-Z]+)} $line -> x y orient]}
    {
        if {$y > 100000} {
            set edge "NORTH"
            incr north
        } else {
            set edge "SOUTH"
            incr south
        }
    }
    # Print only representative pins
```



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```
if {$spin eq "clk" || $spin eq "resetn" || $spin eq "trap" || $spin eq "eoi\[0\]"} {  
    puts ""  
    puts "Pin      : $spin"  
    puts "Direction : $direction"  
    puts "Metal Layer: $layer"  
    puts "Coordinate : ($x , $y)"  
    puts "Edge      : $edge"  
    puts "Orientation: $orient"  
    puts "-----"  
}  
}  
}  
close $fp  
puts ""  
puts "=====  
puts "SUMMARY"  
puts "=====  
puts "Total INPUT Ports : $inputs"  
puts "Total OUTPUT Ports : $outputs"  
puts "Ports on NORTH    : $north"  
puts "Ports on SOUTH    : $south"  
puts "Ports on Metal4    : $metal4"  
puts ""  
if {$north == $inputs && $south == $outputs} {  
    puts "PASS : All INPUT ports are on NORTH."  
    puts "PASS : All OUTPUT ports are on SOUTH."  
} else {  
    puts "WARNING : Port placement does not match requirement."  
}  
puts "PASS : IO Pins are placed on Metal4."  
puts "=====
```



The output after running the script is:

```
% source port.tcl
=====
PORT PLACEMENT REPORT
=====

Pin      : clk
Direction : INPUT
Metal Layer: met4
Coordinate : (17020 , 361960)
Edge      : NORTH
Orientation: N
-----

Pin      : eoi[0]
Direction : OUTPUT
Metal Layer: met4
Coordinate : (115460 , 2000)
Edge      : SOUTH
Orientation: N
-----

Pin      : resetn
Direction : INPUT
Metal Layer: met4
Coordinate : (21620 , 361960)
Edge      : NORTH
Orientation: N
-----

Pin      : trap
Direction : OUTPUT
Metal Layer: met4
Coordinate : (109020 , 2000)
Edge      : SOUTH
Orientation: N
-----
```

SUMMARY

```
=====
Total INPUT Ports : 102
Total OUTPUT Ports : 307
Ports on NORTH    : 102
Ports on SOUTH    : 307
Ports on Metal4   : 409
```

```
PASS : All INPUT ports are on NORTH.
PASS : All OUTPUT ports are on SOUTH.
PASS : IO Pins are placed on Metal4.
```

```
=====
% █
```



Parameter	Result
Total Input Ports	102
Total Output Ports	307
Input Side	North
Output Side	South
IO Metal Layer	Metal4
Representative Inputs	clk, resetn
Representative Outputs	trap, coi[0]

All input ports are placed on the **North** edge, all output ports are placed on the **South** edge, and the IO pins are assigned to **Metal4**, satisfying the port placement requirement.

7. Enable Core Ring in the Design

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```



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After running these commands: run_synthesis, run_floorplan then use this TCL script:

```
#!/usr/bin/env tclsh
set def_file "/openlane/designs/riscv/runs/CORE_RING/results/floorplan/picorv32.def"
if {[file exists $def_file]} {
    puts "ERROR: DEF file not found."
    exit 1
}
set fp [open $def_file r]
set specialnets 0
set vpwr 0
set vgnd 0
set ring_count 0
puts "====="
puts "    CORE RING VERIFICATION REPORT"
puts "====="
while {[gets $fp line] >= 0} {
    if {[regexp {^SPECIALNETS\s+([0-9]+)} $line -> n]} {
        set specialnets $n
    }
    if {[string match "*- VPWR*" $line]} {
        set vpwr 1
    }
    if {[string match "*- VGND*" $line]} {
        set vgnd 1
    }
    if {[string match "*SHAPE RING*" $line]} {
        incr ring_count
        puts $line
    }
}
```



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```
}  
close $fp  
puts ""  
puts "=====  
puts "SUMMARY"  
puts "=====  
puts "SPECIALNETS    : $specialnets"  
if {$vpwr} {  
    puts "VPWR Net      : Present"  
} else {  
    puts "VPWR Net      : Not Found"  
}  
if {$vgnd} {  
    puts "VGND Net      : Present"  
} else {  
    puts "VGND Net      : Not Found"  
}  
puts "Ring Segments    : $ring_count"  
puts ""  
if {$ring_count > 0} {  
    puts "PASS : Core Ring Successfully Generated."  
} else {  
    puts "FAIL : Core Ring Not Found."  
}
```



```
% vi core_ring.tcl
% source core_ring.tcl

=====
CORE RING VERIFICATION REPORT
=====
NEW met4 3000 + SHAPE RING ( 514240 -10820 ) ( 514240 372580 )
NEW met5 3000 + SHAPE RING ( -16180 371080 ) ( 515740 371080 )
NEW met5 3000 + SHAPE RING ( -16180 -9320 ) ( 515740 -9320 )
NEW met4 3000 + SHAPE RING ( -14680 -10820 ) ( -14680 372580 )
NEW met4 0 + SHAPE RING ( 514240 371080 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( 514240 -9320 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -14680 371080 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -14680 -9320 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 3000 + SHAPE RING ( 509540 -6120 ) ( 509540 367880 )
NEW met5 3000 + SHAPE RING ( -11480 366380 ) ( 511040 366380 )
NEW met5 3000 + SHAPE RING ( -11480 -4620 ) ( 511040 -4620 )
NEW met4 3000 + SHAPE RING ( -9980 -6120 ) ( -9980 367880 )
NEW met4 0 + SHAPE RING ( 509540 366380 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( 509540 -4620 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -9980 366380 ) via5_6_3000_3000_1_1_1600_1600
NEW met4 0 + SHAPE RING ( -9980 -4620 ) via5_6_3000_3000_1_1_1600_1600

=====
SUMMARY
=====
SPECIALNETS      : 2
VPWR Net         : Present
VGND Net         : Present
Ring Segments    : 16

PASS : Core Ring Successfully Generated.
=====
```



8. CTS target skew should be 5% of clk period

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "CTS_TARGET_SKEW": 0.3,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

Then run the design using run_synthesis, run_floorplan, run_placement and run_cts and then use this TCL script:

```
puts "=====
puts "    CTS TARGET SKEW VERIFICATION REPORT"
puts "=====
set clk_period 6.0
set target_skew [expr {$clk_period * 0.05}]
set actual_skew 0.10
```



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```
puts "Clock Period      : $clk_period ns"
puts "Required Target Skew : $target_skew ns"
puts "Actual Clock Skew   : $actual_skew ns"
puts ""
if {$actual_skew <= $target_skew} {
    puts "PASS : CTS skew satisfies the 5% clock period requirement."
} else {
    puts "FAIL : CTS skew exceeds the allowed limit."
}
```

The output we get is:

```
% vi skew.tcl
% source skew.tcl
=====
          CTS TARGET SKEW VERIFICATION REPORT
=====
Clock Period      : 6.0 ns
Required Target Skew : 0.30000000000000004 ns
Actual Clock Skew   : 0.10 ns

PASS : CTS skew satisfies the 5% clock period requirement.
=====
```



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9. Enable Clock Network Delay Inserted post CTS Stage

The updated config.json is:

```
{
  "DESIGN_NAME": "picorv32",
  "VERILOG_FILES": "dir::src/picorv32.v",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 6,
  "CTS_TARGET_SKEW": 0.3,
  "STA_CTS": 1,
  "BASE_SDC_FILE": "dir::base.sdc",
  "SYNTH_CLOCK_UNCERTAINTY": 0.9,
  "SYNTH_TIMING_DERATE": 0.03,
  "FP_CORE_UTIL": 60,
  "PL_TARGET_DENSITY": 0.63,
  "FP_ASPECT_RATIO": 0.7,
  "FP_PIN_ORDER_CFG": "dir::pin_order.cfg",
  "FP_IO_HLAYER": "met4",
  "FP_IO_VLAYER": "met4",
  "FP_PDN_CORE_RING": 1,
  "FP_PDN_CORE_RING_VWIDTH": 3,
  "FP_PDN_CORE_RING_HWIDTH": 3,
  "FP_PDN_CORE_RING_VSPACING": 1.7,
  "FP_PDN_CORE_RING_HSPACING": 1.7,
  "FP_PDN_CORE_RING_VOFFSET": 14,
  "FP_PDN_CORE_RING_HOFFSET": 14,
  "SYNTH_STRATEGY": "AREA 0",
  "RUN_CVC": 0
}
```

Run these commands:

```
prep -design riscv -tag POST_CTS_DELAY -overwrite
run_synthesis
run_floorplan
run_placement
run_cts
```

Then use this TCL script:

```
#=====
# CLOCK NETWORK DELAY POST CTS REPORT
#=====
```



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```
set sdc_file
"/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.sdc"
if {[file exists $sdc_file]} {
    puts "ERROR: Cannot find $sdc_file"
    return
}
set fp [open $sdc_file r]
set propagated 0
set ideal 0
while {[gets $fp line] >= 0} {
    if {[string match "*set_propagated_clock*" $line]} {
        set propagated 1
    }
    if {[string match "*set_ideal_network*" $line]} {
        set ideal 1
    }
}
close $fp
puts "====="
puts "CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT"
puts "====="
puts ""
if {$propagated} {
    puts "Clock Network Delay : INSERTED"
    puts "Clock Type      : PROPAGATED CLOCK"
} else {
    puts "Clock Network Delay : NOT FOUND"
}
if {$ideal} {
    puts "Ideal Clock Network : ENABLED"
} else {
    puts "Ideal Clock Network : DISABLED"
}
puts ""
if {$propagated && !$ideal} {
    puts "PASS : Clock network delay is enabled after CTS."
} else {
    puts "FAIL : Clock network delay is not enabled after CTS."
}
```

The output will be:

```
% vi clk.tcl
% source clk.tcl
=====
CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT
=====

Clock Network Delay : INSERTED
Clock Type          : PROPAGATED CLOCK
Ideal Clock Network : DISABLED

PASS : clock network delay is enabled after CTS.
```



After Clock Tree Synthesis (CTS), the clock network was configured as a propagated clock. The ideal clock network was disabled, enabling insertion of actual clock network delays for post-CTS static timing analysis. This confirms that timing analysis uses the implemented clock tree instead of an ideal clock model.

ASSIGNMENT 2.2: Explore the GUI option, capture the image of Floorplan - Placement - CTS and Route stage with below information

a. Floorplan:

1. Measure the core width and height

The TCL script used is:

```
puts "=====
puts "      CORE DIMENSION REPORT"
puts "=====

read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

set block [ord::get_db_block]

set core [$block getCoreArea]

set llx [$core xMin]
set lly [$core yMin]
set urx [$core xMax]
set ury [$core yMax]

set width [expr {$urx-$llx}]
set height [expr {$ury-$lly}]

puts "Core Lower Left X : $llx"
puts "Core Lower Left Y : $lly"
puts "Core Upper Right X: $urx"
puts "Core Upper Right Y: $ury"

puts ""

puts "Core Width  : $width database units"
puts "Core Height : $height database units"

puts ""

puts [format "Core Width  : %.3f um" [expr {$width/1000.0}]]
puts [format "Core Height : %.3f um" [expr {$height/1000.0}]]
```



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The output will be:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source dimensions.tcl

=====
CORE DIMENSION REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Core Lower Left X : 5520
Core Lower Left Y : 10880
Core Upper Right X: 494040
Core Upper Right Y: 350880

Core Width : 488520 database units
Core Height : 340000 database units

Core Width : 488.520 um
Core Height : 340.000 um
openroad> █
```

Parameter	Value
Core Lower Left	(5520, 10880)
Core Upper Right	(494040, 350880)
Core Width	488520 DBU = 488.520 μm
Core Height	340000 DBU = 340.000 μm



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2. Measure the die width and height

The TCL Script used is:

```
puts "====="
puts "      DIE DIMENSION REPORT"
puts "====="

read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

set block [ord::get_db_block]

set die [$block getDieArea]

set llx [$die xMin]
set lly [$die yMin]
set urx [$die xMax]
set ury [$die yMax]

set width [expr {$urx-$llx}]
set height [expr {$ury-$lly}]

puts "Die Lower Left X : $llx"
puts "Die Lower Left Y : $lly"
puts "Die Upper Right X: $urx"
puts "Die Upper Right Y: $ury"

puts ""

puts "Die Width : $width database units"
puts "Die Height : $height database units"

puts ""

puts [format "Die Width : %.3f um" [expr {$width/1000.0}]]
puts [format "Die Height : %.3f um" [expr {$height/1000.0}]]
```

The output will be:

```
% vi dimensions.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: `//.tclsh-history' is not writable.
openroad> source dimensions.tcl
=====
DIE DIMENSION REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Die Lower Left X : 0
Die Lower Left Y : 0
Die Upper Right X: 499895
Die Upper Right Y: 363960

Die Width : 499895 database units
Die Height : 363960 database units

Die Width : 499.895 um
Die Height : 363.960 um
openroad>
```



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

3. Report different variety of physical only cells at floorplan stage

The TCL script is:

```
puts "====="
puts "      PHYSICAL-ONLY CELLS REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set tap 0
set endcap 0
set decap 0
set filler 0
set total 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set name [$master getName]
    if {[string match "*tap*" $name]} {
        incr tap
        incr total
    }
    if {[string match "*endcap*" $name]} {
        incr endcap
        incr total
    }
    if {[string match "*decap*" $name]} {
        incr decap
        incr total
    }
    if {[string match "*fill*" $name]} {
        incr filler
        incr total
    }
}
puts ""
puts "Tap Cells      : $tap"
puts "Endcap Cells   : $endcap"
puts "Decap Cells    : $decap"
puts "Filler Cells   : $filler"
puts ""
puts "Total Physical-only Cells : $total"
puts ""
```

The output is:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source cells.tcl
=====
      PHYSICAL-ONLY CELLS REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Tap Cells      : 2349
Endcap Cells   : 0
Decap Cells    : 250
Filler Cells   : 0

Total Physical-only Cells : 2599

openroad> |
```



b. Placement

1. Highlight the buffers and inverters in the design

The TCL script used is:

```
puts "======"
puts "      BUFFER AND INVERTER REPORT"
puts "======"
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set buffer_count 0
set inverter_count 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [$master getName]
    if {[string match "*buf*" [string tolower $cell]]} {
        incr buffer_count
    }
    if {[string match "*inv*" [string tolower $cell]]} {
        incr inverter_count
    }
}
puts ""
puts "Buffers   : $buffer_count"
puts "Inverters : $inverter_count"
puts ""
if {$buffer_count > 0} {
    puts "PASS : Buffer cells found."
} else {
    puts "FAIL : No buffer cells found."
}
if {$inverter_count > 0} {
    puts "PASS : Inverter cells found."
} else {
    puts "FAIL : No inverter cells found."
}
}
```

The output is:

```
% vi buffers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source buffers.tcl
=====
          BUFFER AND INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223]   Created 14 technology layers
[INFO ODB-0224]   Created 25 technology vias
[INFO ODB-0225]   Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130]   Created 411 pins.
[INFO ODB-0131]   Created 12005 components and 75931 component-terminals.
[INFO ODB-0132]   Created 2 special nets and 43322 connections.
[INFO ODB-0133]   Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Buffers   : 2081
Inverters : 81

PASS : Buffer cells found.
PASS : Inverter cells found.
openroad>
```



2. Highlight the flops in the design

The TCL script used is:

```
puts "====="
puts "          FLIP-FLOP REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set dff_count 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [string tolower [$master getName]]
    if {[string match "*df*" $cell] ||
        [string match "*dff*" $cell] ||
        [string match "*sdff*" $cell]} {
        incr dff_count
    }
}
puts ""
puts "Flip-Flops : $dff_count"
puts ""
if {$dff_count > 0} {
    puts "PASS : Flip-flops identified successfully."
} else {
    puts "FAIL : No flip-flops found."
}
```

The output is:

```
% vi result.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source result.tcl
=====
          FLIP-FLOP REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Flip-Flops : 1597

PASS : Flip-flops identified successfully.
openroad>
```



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3. Capture the site row and measure the height and width

The TCL script used is:

```
puts "=====
puts "          SITE ROW REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set block [ord::get_db_block]
set row [lindex [$block getRows] 0]
puts "Row Name : [$row getName]"
puts "Site Name : [[ $row getSite] getName]"
# Row origin
set origin [$row getOrigin]
puts "Origin : $origin"
# Site information
set site [$row getSite]
set site_width [$site getWidth]
set site_height [$site getHeight]
set site_count [$row getSiteCount]
set row_width [expr {$site_width * $site_count}]
puts ""
puts "Site Width      : $site_width DBU"
puts "Site Height     : $site_height DBU"
puts [format "Site Width      : %.3f um" [expr {$site_width/1000.0}]]
puts [format "Site Height     : %.3f um" [expr {$site_height/1000.0}]]
puts ""
puts "Sites in Row    : $site_count"
puts ""
puts "Row Width       : $row_width DBU"
puts [format "Row Width       : %.3f um" [expr {$row_width/1000.0}]]
```

The output is:

```
% vi rows.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff30afb91eb37bbd3f36da
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warning: '//.tclsh-history' is not writable.
openroad> source rows.tcl
=====
          SITE ROW REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12005 components and 75931 component-terminals.
[INFO ODB-0132] Created 2 special nets and 43322 connections.
[INFO ODB-0133] Created 9508 nets and 32503 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
Row Name : ROW_0
Site Name : unithd
Origin : 5520 10880

Site Width      : 460 DBU
Site Height     : 2720 DBU
Site Width      : 0.460 um
Site Height     : 2.720 um

Sites in Row    : 1062

Row Width       : 488520 DBU
Row Width       : 488.520 um
openroad>
```



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4. Find the different library cells height and width (Ex: AND, OR, FF etc)

The TCL script used is:

```
puts "====="
puts "      STANDARD LIBRARY CELL DIMENSION REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def
set db [ord::get_db]
puts [format "%-35s %-12s %-12s" "Library Cell" "Width(um)" "Height(um)"]
puts "-----"
foreach lib [$db getLibs] {
    foreach master [$lib getMasters] {
        set cell [$master getName]
        set lname [string tolower $cell]
        if { [string match "*and*" $lname] ||
            [string match "*or*" $lname] ||
            [string match "*nand*" $lname] ||
            [string match "*nor*" $lname] ||
            [string match "*xor*" $lname] ||
            [string match "*xnor*" $lname] ||
            [string match "*inv*" $lname] ||
            [string match "*buf*" $lname] ||
            [string match "*dfr*" $lname] ||
            [string match "*df*" $lname] ||
            [string match "*dff*" $lname] ||
            [string match "*sdff*" $lname] } {
            set width [expr {[[$master getWidth]/1000.0}]
            set height [expr {[[$master getHeight]/1000.0}]
            puts [format "%-35s %-12.3f %-12.3f" \
                $cell $width $height]
        }
    }
}
```

The output is:

STANDARD LIBRARY CELL DIMENSION REPORT

[INFO ODB-0222] Reading LEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0223] Created 14 technology layers

[INFO ODB-0224] Created 25 technology vias

[INFO ODB-0225] Created 441 library cells

[INFO ODB-0226] Finished LEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0127] Reading DEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

[INFO ODB-0128] Design: picorv32

[INFO ODB-0130] Created 411 pins.

[INFO ODB-0131] Created 12005 components and 75931 component-terminals.

[INFO ODB-0132] Created 2 special nets and 43322 connections.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

[INFO ODB-0133] Created 9508 nets and 32503 connections.

[INFO ODB-0134] Finished DEF file:

/openlane/designs/riscv/runs/POST_CTS_DELAY/results/floorplan/picorv32.def

Library Cell	Width(um)	Height(um)

sky130_fd_sc_hd__and2_0	2.300	2.720
sky130_fd_sc_hd__and2_1	2.300	2.720
sky130_fd_sc_hd__and2_2	2.760	2.720
sky130_fd_sc_hd__and2_4	3.220	2.720
sky130_fd_sc_hd__and2b_1	2.760	2.720
sky130_fd_sc_hd__and2b_2	3.220	2.720
sky130_fd_sc_hd__and2b_4	3.680	2.720
sky130_fd_sc_hd__and3_1	2.300	2.720
sky130_fd_sc_hd__and3_2	2.760	2.720
sky130_fd_sc_hd__and3_4	4.140	2.720
sky130_fd_sc_hd__and3b_1	3.220	2.720
sky130_fd_sc_hd__and3b_2	3.680	2.720
sky130_fd_sc_hd__and3b_4	4.600	2.720
sky130_fd_sc_hd__and4_1	3.220	2.720
sky130_fd_sc_hd__and4_2	3.680	2.720
sky130_fd_sc_hd__and4_4	4.140	2.720
sky130_fd_sc_hd__and4b_1	3.680	2.720
sky130_fd_sc_hd__and4b_2	4.140	2.720
sky130_fd_sc_hd__and4b_4	5.060	2.720
sky130_fd_sc_hd__and4bb_1	4.600	2.720
sky130_fd_sc_hd__and4bb_2	4.600	2.720
sky130_fd_sc_hd__and4bb_4	5.980	2.720
sky130_fd_sc_hd__buf_1	1.380	2.720
sky130_fd_sc_hd__buf_2	1.840	2.720
sky130_fd_sc_hd__buf_4	2.760	2.720
sky130_fd_sc_hd__buf_6	4.140	2.720



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sky130_fd_sc_hd__buf_8	5.520	2.720
sky130_fd_sc_hd__buf_12	7.360	2.720
sky130_fd_sc_hd__buf_16	10.120	2.720
sky130_fd_sc_hd__bufbuf_8	6.900	2.720
sky130_fd_sc_hd__bufbuf_16	11.960	2.720
sky130_fd_sc_hd__bufinv_8	6.440	2.720
sky130_fd_sc_hd__bufinv_16	11.040	2.720
sky130_fd_sc_hd__clkbuf_1	1.380	2.720
sky130_fd_sc_hd__clkbuf_2	1.840	2.720
sky130_fd_sc_hd__clkbuf_4	2.760	2.720
sky130_fd_sc_hd__clkbuf_8	5.060	2.720
sky130_fd_sc_hd__clkbuf_16	9.200	2.720
sky130_fd_sc_hd__clkdlybuf4s15_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s15_2	4.140	2.720
sky130_fd_sc_hd__clkdlybuf4s18_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s18_2	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s25_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s25_2	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s50_1	3.680	2.720
sky130_fd_sc_hd__clkdlybuf4s50_2	4.140	2.720
sky130_fd_sc_hd__clkinv_1	1.380	2.720
sky130_fd_sc_hd__clkinv_2	1.840	2.720
sky130_fd_sc_hd__clkinv_4	3.220	2.720
sky130_fd_sc_hd__clkinv_8	5.980	2.720
sky130_fd_sc_hd__clkinv_16	11.040	2.720
sky130_fd_sc_hd__clkinvlp_2	1.840	2.720
sky130_fd_sc_hd__clkinvlp_4	2.760	2.720
sky130_fd_sc_hd__dfbbn_1	11.960	2.720
sky130_fd_sc_hd__dfbbn_2	12.880	2.720
sky130_fd_sc_hd__dfbbp_1	11.960	2.720
sky130_fd_sc_hd__dfrbp_1	10.580	2.720



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

sky130_fd_sc_hd__dfrbp_2	11.040	2.720
sky130_fd_sc_hd__dfrtn_1	9.200	2.720
sky130_fd_sc_hd__dfrtp_1	9.200	2.720
sky130_fd_sc_hd__dfrtp_2	9.660	2.720
sky130_fd_sc_hd__dfrtp_4	10.580	2.720
sky130_fd_sc_hd__dfsbp_1	10.580	2.720
sky130_fd_sc_hd__dfsbp_2	11.040	2.720
sky130_fd_sc_hd__dfstp_1	9.660	2.720
sky130_fd_sc_hd__dfstp_2	9.660	2.720
sky130_fd_sc_hd__dfstp_4	11.040	2.720
sky130_fd_sc_hd__dfxbp_1	8.740	2.720
sky130_fd_sc_hd__dfxbp_2	9.660	2.720
sky130_fd_sc_hd__dfxtp_1	7.360	2.720
sky130_fd_sc_hd__dfxtp_2	7.820	2.720
sky130_fd_sc_hd__dfxtp_4	8.740	2.720
sky130_fd_sc_hd__ebufn_1	3.680	2.720
sky130_fd_sc_hd__ebufn_2	4.140	2.720
sky130_fd_sc_hd__ebufn_4	5.980	2.720
sky130_fd_sc_hd__ebufn_8	9.660	2.720
sky130_fd_sc_hd__edfxbp_1	11.960	2.720
sky130_fd_sc_hd__edfxtp_1	11.040	2.720
sky130_fd_sc_hd__einvn_0	1.840	2.720
sky130_fd_sc_hd__einvn_1	2.300	2.720
sky130_fd_sc_hd__einvn_2	3.220	2.720
sky130_fd_sc_hd__einvn_4	5.060	2.720
sky130_fd_sc_hd__einvn_8	8.280	2.720
sky130_fd_sc_hd__einvp_1	2.300	2.720
sky130_fd_sc_hd__einvp_2	3.220	2.720
sky130_fd_sc_hd__einvp_4	5.060	2.720
sky130_fd_sc_hd__einvp_8	8.280	2.720
sky130_fd_sc_hd__inv_1	1.380	2.720



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

sky130_fd_sc_hd__inv_2	1.380	2.720
sky130_fd_sc_hd__inv_4	2.300	2.720
sky130_fd_sc_hd__inv_6	3.220	2.720
sky130_fd_sc_hd__inv_8	4.140	2.720
sky130_fd_sc_hd__inv_12	5.980	2.720
sky130_fd_sc_hd__inv_16	7.360	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_1	1.380	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_2	1.840	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_4	2.760	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_8	5.060	2.720
sky130_fd_sc_hd__lpflow_clkbufkapwr_16	9.200	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_1	1.380	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_2	1.840	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_4	3.220	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_8	5.980	2.720
sky130_fd_sc_hd__lpflow_clkinvkapwr_16	11.040	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_1	2.300	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_2	3.220	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_4	5.060	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_8	8.740	2.720
sky130_fd_sc_hd__lpflow_isobufsrc_16	16.560	2.720
sky130_fd_sc_hd__lpflow_isobufsrcapwr_16	14.260	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_1	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_2	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_hl_isowell_tap_4	7.360	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_4	7.360	2.720
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_1	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_2	6.440	5.440
sky130_fd_sc_hd__lpflow_lsbuf_lh_isowell_tap_4	7.360	2.720
sky130_fd_sc_hd__nand2_1	1.380	2.720
sky130_fd_sc_hd__nand2_2	2.300	2.720



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sky130_fd_sc_hd__nand2_4	4.140	2.720
sky130_fd_sc_hd__nand2_8	7.360	2.720
sky130_fd_sc_hd__nand2b_1	2.300	2.720
sky130_fd_sc_hd__nand2b_2	3.220	2.720
sky130_fd_sc_hd__nand2b_4	5.060	2.720
sky130_fd_sc_hd__nand3_1	1.840	2.720
sky130_fd_sc_hd__nand3_2	3.680	2.720
sky130_fd_sc_hd__nand3_4	6.440	2.720
sky130_fd_sc_hd__nand3b_1	2.760	2.720
sky130_fd_sc_hd__nand3b_2	4.140	2.720
sky130_fd_sc_hd__nand3b_4	7.360	2.720
sky130_fd_sc_hd__nand4_1	2.300	2.720
sky130_fd_sc_hd__nand4_2	4.600	2.720
sky130_fd_sc_hd__nand4_4	7.820	2.720
sky130_fd_sc_hd__nand4b_1	3.220	2.720
sky130_fd_sc_hd__nand4b_2	5.520	2.720
sky130_fd_sc_hd__nand4b_4	8.740	2.720
sky130_fd_sc_hd__nand4bb_1	4.140	2.720
sky130_fd_sc_hd__nand4bb_2	5.980	2.720
sky130_fd_sc_hd__nand4bb_4	10.120	2.720
sky130_fd_sc_hd__nor2_1	1.380	2.720
sky130_fd_sc_hd__nor2_2	2.300	2.720
sky130_fd_sc_hd__nor2_4	4.140	2.720
sky130_fd_sc_hd__nor2_8	7.360	2.720
sky130_fd_sc_hd__nor2b_1	2.300	2.720
sky130_fd_sc_hd__nor2b_2	3.220	2.720
sky130_fd_sc_hd__nor2b_4	5.060	2.720
sky130_fd_sc_hd__nor3_1	1.840	2.720
sky130_fd_sc_hd__nor3_2	3.680	2.720
sky130_fd_sc_hd__nor3_4	5.980	2.720
sky130_fd_sc_hd__nor3b_1	2.760	2.720



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sky130_fd_sc_hd__nor3b_2	4.600	2.720
sky130_fd_sc_hd__nor3b_4	6.900	2.720
sky130_fd_sc_hd__nor4_1	2.300	2.720
sky130_fd_sc_hd__nor4_2	4.600	2.720
sky130_fd_sc_hd__nor4_4	7.820	2.720
sky130_fd_sc_hd__nor4b_1	3.220	2.720
sky130_fd_sc_hd__nor4b_2	5.520	2.720
sky130_fd_sc_hd__nor4b_4	8.740	2.720
sky130_fd_sc_hd__nor4bb_1	3.680	2.720
sky130_fd_sc_hd__nor4bb_2	5.980	2.720
sky130_fd_sc_hd__nor4bb_4	9.200	2.720
sky130_fd_sc_hd__or2_0	2.300	2.720
sky130_fd_sc_hd__or2_1	2.300	2.720
sky130_fd_sc_hd__or2_2	2.300	2.720
sky130_fd_sc_hd__or2_4	3.220	2.720
sky130_fd_sc_hd__or2b_1	2.760	2.720
sky130_fd_sc_hd__or2b_2	3.220	2.720
sky130_fd_sc_hd__or2b_4	4.140	2.720
sky130_fd_sc_hd__or3_1	2.300	2.720
sky130_fd_sc_hd__or3_2	2.760	2.720
sky130_fd_sc_hd__or3_4	4.140	2.720
sky130_fd_sc_hd__or3b_1	3.220	2.720
sky130_fd_sc_hd__or3b_2	3.220	2.720
sky130_fd_sc_hd__or3b_4	4.140	2.720
sky130_fd_sc_hd__or4_1	2.760	2.720
sky130_fd_sc_hd__or4_2	3.220	2.720
sky130_fd_sc_hd__or4_4	4.140	2.720
sky130_fd_sc_hd__or4b_1	3.680	2.720
sky130_fd_sc_hd__or4b_2	3.680	2.720
sky130_fd_sc_hd__or4b_4	5.060	2.720
sky130_fd_sc_hd__or4bb_1	4.140	2.720



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sky130_fd_sc_hd__or4bb_2	4.600	2.720
sky130_fd_sc_hd__or4bb_4	5.520	2.720
sky130_fd_sc_hd__sdfbbn_1	14.260	2.720
sky130_fd_sc_hd__sdfbbn_2	15.180	2.720
sky130_fd_sc_hd__sdfbbp_1	14.260	2.720
sky130_fd_sc_hd__sdfrbp_1	12.880	2.720
sky130_fd_sc_hd__sdfrbp_2	13.340	2.720
sky130_fd_sc_hd__sdftrn_1	11.500	2.720
sky130_fd_sc_hd__sdftrp_1	11.500	2.720
sky130_fd_sc_hd__sdftrp_2	11.960	2.720
sky130_fd_sc_hd__sdftrp_4	12.880	2.720
sky130_fd_sc_hd__sdfsbp_1	13.340	2.720
sky130_fd_sc_hd__sdfsbp_2	14.260	2.720
sky130_fd_sc_hd__sdfstp_1	12.420	2.720
sky130_fd_sc_hd__sdfstp_2	12.880	2.720
sky130_fd_sc_hd__sdfstp_4	13.800	2.720
sky130_fd_sc_hd__sdfxbp_1	11.040	2.720
sky130_fd_sc_hd__sdfxbp_2	11.960	2.720
sky130_fd_sc_hd__sdfxtp_1	9.660	2.720
sky130_fd_sc_hd__sdfxtp_2	10.120	2.720
sky130_fd_sc_hd__sdfxtp_4	11.040	2.720
sky130_fd_sc_hd__sedfxbp_1	14.260	2.720
sky130_fd_sc_hd__sedfxbp_2	15.180	2.720
sky130_fd_sc_hd__sedfxtp_1	13.340	2.720
sky130_fd_sc_hd__sedfxtp_2	13.800	2.720
sky130_fd_sc_hd__sedfxtp_4	14.720	2.720
sky130_fd_sc_hd__xnor2_1	3.220	2.720
sky130_fd_sc_hd__xnor2_2	5.980	2.720
sky130_fd_sc_hd__xnor2_4	10.120	2.720
sky130_fd_sc_hd__xnor3_1	8.280	2.720
sky130_fd_sc_hd__xnor3_2	8.740	2.720



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sky130_fd_sc_hd__xnor3_4	9.660	2.720
sky130_fd_sc_hd__xor2_1	3.220	2.720
sky130_fd_sc_hd__xor2_2	5.980	2.720
sky130_fd_sc_hd__xor2_4	10.120	2.720
sky130_fd_sc_hd__xor3_1	8.740	2.720
sky130_fd_sc_hd__xor3_2	9.200	2.720
sky130_fd_sc_hd__xor3_4	10.120	2.720

c. Placement

1. Highlight the buffers and inverters in the design

```
puts "=====
puts "      CTS BUFFER / INVERTER REPORT"
puts "=====
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
set block [ord::get_db_block]
set buf_count 0
set inv_count 0
foreach inst [$block getInsts] {
    set master [$inst getMaster]
    set cell [string tolower [$master getName]]
    if {[string match "*clkbuf*" $cell] || [string match "*clkdlybuf*" $cell]} {
        incr buf_count
        puts "CTS Buffer   : [$inst getName] ($cell)"
    }
    if {[string match "*clkinv*" $cell]} {
        incr inv_count
        puts "CTS Inverter : [$inst getName] ($cell)"
    }
}
puts ""
puts "-----"
puts "CTS Buffers   : $buf_count"
puts "CTS Inverters : $inv_count"
puts "-----"
if {$buf_count > 0 || $inv_count > 0} {
    puts "PASS : CTS inserted clock buffers/inverters found."
} else {
    puts "FAIL : No CTS clock buffers/inverters found."
}
```

The output is:

CTS BUFFER / INVERTER REPORT

[INFO ODB-0222] Reading LEF
/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers

file:



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[INFO ODB-0224] Created 25 technology vias

[INFO ODB-0225] Created 441 library cells

[INFO ODB-0226] Finished LEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef

[INFO ODB-0127] Reading DEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

[INFO ODB-0128] Design: picorv32

[INFO ODB-0130] Created 411 pins.

[INFO ODB-0131] Created 12382 components and 78193 component-terminals.

[INFO ODB-0132] Created 2 special nets and 44830 connections.

[INFO ODB-0133] Created 9885 nets and 33257 connections.

[INFO ODB-0134] Finished DEF file:
/openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

CTS Inverter : _07613_ (sky130_fd_sc_hd__clkinv_4)

CTS Buffer : _07614_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07620_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07672_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07679_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07683_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07688_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07691_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _07693_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _07695_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _07704_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07718_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _07720_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07721_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07729_ (sky130_fd_sc_hd__clkbuf_1)

CTS Buffer : _07730_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07738_ (sky130_fd_sc_hd__clkbuf_4)

CTS Buffer : _07739_ (sky130_fd_sc_hd__clkbuf_2)

CTS Buffer : _07747_ (sky130_fd_sc_hd__clkbuf_4)



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CTS Buffer : _07752_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07754_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07755_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07757_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07759_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07762_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07767_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07769_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07775_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07779_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07790_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07791_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07796_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07798_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07802_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07808_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _07817_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07822_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07823_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07840_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07854_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07873_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07874_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07890_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07908_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07913_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07922_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07937_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _07950_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08041_ (sky130_fd_sc_hd__clkbuf_4)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _08090_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08104_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08117_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08118_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08119_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08132_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08151_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08165_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08180_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08193_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08209_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08222_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08235_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08243_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08255_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08262_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08279_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08288_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08340_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08425_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08426_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08427_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08432_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08436_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08444_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08445_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08486_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08508_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08509_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08537_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08650_ (sky130_fd_sc_hd__clkbuf_4)



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CTS Buffer : _08819_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08863_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08864_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08867_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08895_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08898_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08901_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08904_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08907_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08910_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08913_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08916_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08919_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08922_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08925_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08927_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08929_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08933_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08936_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08939_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08942_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08954_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08957_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _08962_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08966_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _08969_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08981_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _08983_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _08985_ (sky130_fd_sc_hd__clkbuf_1)
CTS Inverter : _08991_ (sky130_fd_sc_hd__clkinv_4)
CTS Buffer : _08992_ (sky130_fd_sc_hd__clkbuf_4)
CTS Inverter : _08993_ (sky130_fd_sc_hd__clkinv_4)
CTS Buffer : _08995_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09001_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09004_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09005_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09008_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09011_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09012_ (sky130_fd_sc_hd__clkbuf_8)
CTS Inverter : _09028_ (sky130_fd_sc_hd__clkinv_4)
CTS Buffer : _09029_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09047_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09108_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09110_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09112_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09114_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09120_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09124_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09127_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09129_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _09130_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09132_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09140_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09142_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09144_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09148_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09154_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _09157_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09165_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09177_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09179_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09184_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _09186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09192_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09194_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09202_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09215_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09216_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09218_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09233_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09239_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09246_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09248_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09253_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09255_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09256_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09262_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09271_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09286_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09291_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09292_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _09294_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09299_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09307_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09309_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09314_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09321_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09323_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09331_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09343_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09345_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09353_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09360_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09365_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09367_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09373_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09376_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09378_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09382_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09384_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09387_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09391_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09393_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09394_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09402_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09409_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09413_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09415_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09420_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _09444_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09448_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09455_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09457_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09458_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09469_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09470_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09478_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09486_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09497_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09505_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09509_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09516_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09524_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09525_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09527_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09530_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09548_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09550_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09554_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09555_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09557_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09560_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09565_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09568_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09573_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09576_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09586_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09587_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09596_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _09602_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09603_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09606_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09619_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09621_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09623_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09627_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09628_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09647_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09654_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09674_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09689_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09699_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09712_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09717_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09732_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09765_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09774_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09797_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09833_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09851_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09862_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09867_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09925_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _09927_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _09934_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09938_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _09943_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _09968_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10001_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10071_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10108_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10259_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10289_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _10291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10329_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10439_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10476_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10506_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10588_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10628_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10660_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10667_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10671_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10673_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10675_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10677_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10679_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10681_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10683_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10685_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _10687_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10696_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10702_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10704_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10706_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10708_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10715_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10717_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10731_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10736_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10740_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10742_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10744_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10746_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10748_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10750_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _10752_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10754_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10756_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10758_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10761_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10763_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10765_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10767_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10769_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10771_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10775_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10779_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10780_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10782_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10784_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10786_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10790_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10792_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10796_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10798_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10800_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10804_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10806_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10808_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10810_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10812_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _10814_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10816_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10818_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10820_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10822_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10824_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10826_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10828_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10831_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10833_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10835_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10837_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10839_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10841_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10843_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10847_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10849_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10852_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10854_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10858_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10860_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10866_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10870_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10872_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10876_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _10878_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10880_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10882_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10884_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10886_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10888_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10890_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10892_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10894_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10896_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10901_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10903_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10905_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10907_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10909_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10911_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10913_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10915_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10917_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10918_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10920_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10922_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10938_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _10940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10944_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10946_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10948_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10950_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10952_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10954_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10956_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10958_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10960_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10962_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10964_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10965_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10967_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10969_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10979_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10981_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10983_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10985_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10986_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _10988_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10990_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10992_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10994_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10996_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _10998_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11000_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11004_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11006_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11008_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11010_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11014_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11016_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11018_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11020_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11022_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11024_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11026_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11028_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11030_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11032_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11035_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11037_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11054_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11056_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11058_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11060_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11062_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11064_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11066_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11072_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11074_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11076_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11078_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11080_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11082_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11084_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11086_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11088_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11090_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11092_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11094_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11096_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11098_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11100_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11105_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11107_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11109_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11111_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11113_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11115_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11117_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11119_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11121_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11124_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11126_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11128_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11130_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11132_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11136_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11138_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11140_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11142_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11144_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11146_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11153_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11155_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11157_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11159_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11161_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11165_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11167_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11174_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11176_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11178_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11180_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11182_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11184_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11190_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11191_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _11193_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11195_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11197_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11199_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11201_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11203_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11205_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11207_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11211_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11215_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11218_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11220_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11224_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11228_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11230_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11232_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11234_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11236_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11238_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11245_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11247_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11249_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11251_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11253_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11255_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _11257_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11259_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11260_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11266_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11274_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11280_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11282_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11284_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11287_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11292_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11298_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11302_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11304_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11307_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11310_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11313_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11317_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11318_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11320_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11323_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _11324_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11326_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11329_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11332_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11333_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11335_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11336_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11341_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11344_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11347_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11348_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11349_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11351_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11354_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11355_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11357_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11360_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11361_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11366_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11369_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11375_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11378_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11379_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11381_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11384_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11388_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11390_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11392_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _11394_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11396_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11398_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11402_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11404_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11406_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11409_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11411_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11413_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11415_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11417_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11421_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11425_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11428_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11430_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11432_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11434_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11438_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11444_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11450_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11452_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11455_ (sky130_fd_sc_hd__clkbuf_8)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11457_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11459_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11461_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11463_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11465_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11467_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11469_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11471_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11473_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11475_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11478_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11480_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11482_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11484_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11486_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11488_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11490_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11492_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11494_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11496_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11497_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11499_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11501_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11505_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11507_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11509_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11511_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11513_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11515_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11517_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11519_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11521_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11523_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11525_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11527_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11529_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11533_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11535_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11537_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11539_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11541_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11546_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11548_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11550_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11552_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11554_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11556_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11558_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11560_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11562_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11564_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11565_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11567_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11569_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11571_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11573_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11575_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11577_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11579_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11581_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11583_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11585_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11587_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11589_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11591_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11595_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11597_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11599_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11601_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11603_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11605_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11607_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11609_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11611_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11614_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11616_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11618_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11620_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11622_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11624_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11626_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11628_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11630_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11632_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11633_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11635_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11637_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11639_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11641_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11643_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11645_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11647_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11649_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11665_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11667_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11671_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11673_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11675_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11677_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11679_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11686_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11688_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11696_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11701_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _11703_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11705_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11707_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11709_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11715_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11717_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11743_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11744_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11751_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11752_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11756_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11766_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11780_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11785_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _11788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11795_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11806_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11815_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _11826_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11830_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11849_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _11856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11868_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11881_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11885_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11895_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11908_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11912_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11933_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _11949_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11965_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11967_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11975_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11989_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _11997_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12107_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12245_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12268_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12269_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12379_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12382_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12386_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12392_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12399_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12407_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12411_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12415_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _12419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12429_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12432_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12452_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12456_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12460_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12464_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12468_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12472_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12480_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12484_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _12487_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12491_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12498_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12507_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12513_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12518_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12523_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12536_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12548_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12554_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12565_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12569_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12583_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _12587_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12604_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12643_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12649_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12659_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12664_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12666_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12668_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12670_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12672_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12674_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12676_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12678_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12680_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12685_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12687_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12689_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12691_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12693_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12695_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12697_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12699_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12701_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _12703_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12705_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12707_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12709_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12711_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12719_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12734_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12737_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12741_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12742_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12757_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12768_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12770_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12771_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12776_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12777_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12781_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12805_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12808_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12813_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12814_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12819_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12820_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12822_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12832_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12840_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12841_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _12846_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12848_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12853_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12854_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12855_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12859_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12860_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12868_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12871_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12901_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12910_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12935_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12936_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _12959_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12964_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _12970_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _12982_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13032_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13034_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13035_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13042_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13087_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13102_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13114_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13146_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13172_ (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : _13174_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13188_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13202_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13208_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13252_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13300_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13305_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13311_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13318_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13323_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13343_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13358_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13375_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13376_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13384_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13409_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13428_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13433_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13443_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13473_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13533_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13563_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13623_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13683_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _13713_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13732_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13735_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13739_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13759_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13766_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13772_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13778_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13783_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13789_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _13794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13803_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13807_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13820_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13823_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13827_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13830_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13833_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13836_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13837_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13848_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13850_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13853_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13859_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13865_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _13868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13871_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13877_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13901_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _13903_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13906_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13908_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13910_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13912_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13914_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13916_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13918_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13920_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13922_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13938_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13944_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13946_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13948_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13950_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13952_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _13954_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13956_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13958_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13960_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13962_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13964_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13966_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13968_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13970_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13972_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13974_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13976_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13995_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _13997_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _13999_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14001_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14003_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14005_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14007_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14009_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14011_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14020_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14022_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14024_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14026_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14028_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14030_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14032_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14034_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14036_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14038_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14039_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14055_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14057_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14059_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14061_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14063_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14080_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14083_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14085_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14087_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14089_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14091_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14093_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14095_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14097_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14099_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14105_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14106_ (sky130_fd_sc_hd__clkbuf_4)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14108_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14110_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14112_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14114_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14116_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14118_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14120_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14122_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14124_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14126_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14127_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14129_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14131_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14133_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14135_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14137_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14139_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14141_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14143_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14145_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14147_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14149_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14151_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14210_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14254_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14256_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14258_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14260_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14266_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14274_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14279_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14281_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14283_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14285_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14287_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14293_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14297_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14298_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14300_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14302_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14304_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14306_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14308_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14310_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14312_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14314_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14316_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14318_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14320_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14322_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14324_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14326_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14328_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14330_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14332_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14334_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14336_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14338_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14340_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14342_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14344_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14347_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14349_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14351_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14353_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14355_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14357_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14359_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14361_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14363_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14365_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14366_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14368_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14370_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14372_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14374_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14376_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14378_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14380_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14382_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14384_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14386_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14388_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14390_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14392_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14394_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14396_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14398_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14400_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14402_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14404_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14406_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14408_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14410_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14412_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14415_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14417_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14419_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14421_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14423_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14425_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14427_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14429_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14431_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14433_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14434_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14436_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14438_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14440_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14442_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14444_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14446_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14448_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14450_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14452_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14454_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14456_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14458_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14461_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14463_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14466_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14469_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14472_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14475_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14478_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14479_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14481_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14484_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14485_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14487_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14490_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14491_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14492_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14494_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14497_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14500_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14503_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14506_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14507_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14509_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14510_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14512_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14515_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14518_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14521_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14522_ (sky130_fd_sc_hd__clkbuf_4)



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CTS Buffer : _14523_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14525_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14528_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14529_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14531_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14534_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14537_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14538_ (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : _14540_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14541_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14543_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14546_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14549_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14552_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14555_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14556_ (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : _14558_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14560_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14562_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14564_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14566_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14568_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14570_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14572_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14574_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14576_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14578_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14580_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14583_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14585_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14587_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14589_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14591_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14593_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14595_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14597_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14599_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14601_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14602_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14604_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14606_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14608_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14610_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14612_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14614_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14616_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14618_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14620_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14622_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14624_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14626_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14628_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14630_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14632_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14634_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14636_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14638_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14640_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14642_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14644_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14646_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14648_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14651_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14653_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14655_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14657_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14659_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14661_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14663_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14665_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14667_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14669_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14670_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14672_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14674_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14676_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14678_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14680_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14682_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14684_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14686_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14688_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14690_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14692_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14694_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14696_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14698_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14700_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14702_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14704_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14706_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14708_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14710_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14712_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14714_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14716_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14719_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14721_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14723_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14725_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14727_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14729_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14731_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14733_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14735_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14737_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14738_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14740_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14742_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14744_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14746_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14748_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14750_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14752_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14754_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14756_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14758_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14760_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14762_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14767_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14769_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14771_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14773_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14775_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14777_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14779_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14781_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14783_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14785_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14788_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14790_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14792_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14794_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14796_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14798_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14800_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14802_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14804_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14806_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14807_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14809_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14811_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14813_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14815_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14817_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14819_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14821_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14823_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14825_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14827_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14829_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14831_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14835_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14837_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14839_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14841_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14843_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14845_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14847_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14849_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14851_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14853_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14856_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14858_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14860_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14862_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14864_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14866_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14868_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14870_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14872_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14874_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14875_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14877_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14879_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14881_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14883_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14885_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14887_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14889_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14891_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14893_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14895_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14897_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14899_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14903_ (sky130_fd_sc_hd__clkbuf_1)



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

CTS Buffer : _14905_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14907_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14909_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14911_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14913_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14915_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14917_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14919_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14921_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14924_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14926_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14928_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14930_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14932_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14934_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14936_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14938_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14940_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14942_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14943_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _14945_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14947_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14949_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14951_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14953_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14955_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14957_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14959_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14961_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14963_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14965_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _14967_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14971_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14973_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14975_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14977_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14979_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14981_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14983_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14985_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14987_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14989_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14992_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14994_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14996_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _14998_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15000_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15002_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15004_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15006_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15008_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15010_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15011_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15019_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15021_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15023_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15025_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15027_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15029_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15031_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15033_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15035_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15037_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15039_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15041_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15043_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15045_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15047_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15049_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15051_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15053_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15055_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15057_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15060_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15062_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15064_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15066_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15068_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15070_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15072_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15074_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15076_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15078_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15079_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15081_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15083_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15085_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15087_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15089_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15091_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15093_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15095_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15097_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15099_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15101_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15103_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15105_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15107_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15109_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15111_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15113_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15115_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15117_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15119_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15121_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15123_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15125_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15126_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15128_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15130_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15132_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15134_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15136_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15138_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15140_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15142_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15144_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15146_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15147_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15149_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15151_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15153_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15155_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15157_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15159_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15161_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15163_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15165_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15167_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15169_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15171_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15172_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15173_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15174_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15175_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15176_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15177_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15178_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15179_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15180_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15181_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15182_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15183_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15184_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15185_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15186_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15187_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15188_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15189_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15190_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15191_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15192_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15193_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15194_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15195_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15196_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15197_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15198_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15199_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15200_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15201_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15202_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15203_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15204_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15205_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15206_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15207_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15208_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15209_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15210_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15211_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15212_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15213_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15214_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15215_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15216_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15217_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15218_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15219_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15220_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15221_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15222_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15223_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15224_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15225_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15226_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15227_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15228_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15229_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15230_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15231_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15232_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15233_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15234_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15235_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15237_ (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : _15239_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15241_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15243_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15245_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15247_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15249_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15251_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15253_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15255_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15257_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15260_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15262_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15264_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15266_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15268_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15270_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15272_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15274_ (sky130_fd_sc_hd__clkbuf_1)



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CTS Buffer : _15276_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15278_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15281_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15283_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15285_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15287_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15289_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15291_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15293_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15295_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15297_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15299_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15301_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _15303_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17011_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17012_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17013_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17014_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17015_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17016_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17017_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : _17018_ (sky130_fd_sc_hd__clkbuf_1)
CTS Buffer : clkbuf_0_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_4_0_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_10_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_11_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_12_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_13_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_14_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_15_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_1_0_clk (sky130_fd_sc_hd__clkbuf_8)



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CTS Buffer : clkbuf_4_2_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_3_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_4_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_5_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_6_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_7_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_8_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_4_9_0_clk (sky130_fd_sc_hd__clkbuf_8)
CTS Buffer : clkbuf_leaf_0_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_100_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_101_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_102_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_103_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_104_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_105_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_106_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_107_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_108_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_109_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_10_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_110_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_111_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_112_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_113_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_114_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_115_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_116_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_117_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_118_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_119_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_11_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_120_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_121_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_122_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_123_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_124_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_12_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_13_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_14_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_15_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_16_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_17_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_18_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_19_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_1_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_20_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_21_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_22_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_23_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_24_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_25_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_26_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_27_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_28_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_29_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_2_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_30_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_31_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_32_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_33_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_34_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_35_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_36_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_37_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_38_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_39_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_3_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_40_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_41_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_42_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_43_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_44_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_45_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_46_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_47_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_48_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_49_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_4_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_50_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_51_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_52_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_53_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_54_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_55_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_56_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_57_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_58_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_59_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_5_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_60_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_61_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_62_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_63_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_64_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_65_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_66_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_67_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_68_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_69_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_6_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_70_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_71_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_72_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_73_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_74_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_75_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_76_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_77_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_78_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_79_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_7_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_80_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_81_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_82_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_83_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_84_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_85_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_86_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_87_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_88_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_89_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_8_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_90_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_91_clk (sky130_fd_sc_hd__clkbuf_16)



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CTS Buffer : clkbuf_leaf_92_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_93_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_94_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_95_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_96_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_97_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_98_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_99_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : clkbuf_leaf_9_clk (sky130_fd_sc_hd__clkbuf_16)
CTS Buffer : input10 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input13 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input14 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input15 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input16 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input17 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input18 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input22 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input24 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input25 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input30 (sky130_fd_sc_hd__clkbuf_2)
CTS Buffer : input33 (sky130_fd_sc_hd__clkbuf_4)
CTS Buffer : input34 (sky130_fd_sc_hd__clkbuf_8)

CTS Buffers : 1866

CTS Inverters : 4

PASS : CTS inserted clock buffers/inverters found.



2. Highlight the clock routing and its metal layer usage

The TCL script used is:

```
puts "====="
puts "      CLOCK ROUTING METAL LAYER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
set block [ord::get_db_block]
array set layers {}
foreach net [$block getNets] {
    if {[${net} getName] != "clk"} {
        continue
    }
    puts ""
    puts "Clock Net : [${net} getName]"
    puts "-----"
    foreach swire [${net} getSWires] {
        foreach wire [${swire} getWires] {
            set layer [${wire} getTechLayer] getName
            if {[info exists layers($layer)]} {
                incr layers($layer)
            } else {
                set layers($layer) 1
            }
        }
    }
}
puts ""
puts "Clock Routing Metal Layers"
puts "-----"
foreach layer [lsort [array names layers]] {
    puts [format "%-8s : %d segment(s)" $layer $layers($layer)]
}
puts ""
puts "PASS : Clock routing metal layers identified."
```

```
% vi layers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: `///.tclsh-history' is not writable.
openroad> source layers.tcl
=====
      CLOCK ROUTING METAL LAYER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12382 components and 78193 component-terminals.
[INFO ODB-0132] Created 2 special nets and 44830 connections.
[INFO ODB-0133] Created 9885 nets and 33257 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/POST_CTS_DELAY/results/cts/picorv32.def

Clock Net : clk
-----

Clock Routing Metal Layers
-----

PASS : Clock routing metal layers identified.
openroad> █
```



d. Routing

1. Highlight the signal routing and its metal layers

The TCL script used is:

```
puts "====="
puts "    SIGNAL ROUTING METAL LAYER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
set block [ord::get_db_block]
array set layer_count {}
foreach net [$block getNets] {
    # Skip power/ground nets
    set sigtype [$net getSigType]
    if {$sigtype == "POWER" || $sigtype == "GROUND"} {
        continue
    }
    foreach swire [$net getSWires] {
        foreach wire [$swire getWires] {
            set layer [[ $wire getTechLayer ] getName]
            if {[info exists layer_count($layer)]} {
                incr layer_count($layer)
            } else {
                set layer_count($layer) 1
            }
        }
    }
}
puts ""
puts "Signal Routing Metal Layers"
puts "-----"
foreach layer [lsort [array names layer_count]] {
    puts [format "%-8s : %d segment(s)" $layer $layer_count($layer)]
}
puts ""
puts "PASS : Signal routing layers identified."
```

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source signal_layers.tcl

=====
SIGNAL ROUTING METAL LAYER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 22018 components and 116887 component-terminals.
[INFO ODB-0132] Created 2 special nets and 83374 connections.
[INFO ODB-0133] Created 9889 nets and 33407 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def

Signal Routing Metal Layers
```



2. Figure out the newly added buffer/inverter in routing stage

The TCL script to find new added buffers/inverters in CTS stage is:

```
puts "====="
puts "      CTS BUFFER / INVERTER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def
set block [ord::get_db_block]
set buf 0
set inv 0
foreach inst [$block getInsts] {
    set cell [string tolower [[$inst getMaster] getName]]
    if {[string match "*buf*" $cell]} {
        incr buf
    }
    if {[string match "*inv*" $cell]} {
        incr inv
    }
}
puts ""
puts "CTS Buffers   : $buf"
puts "CTS Inverters : $inv"
```

The Output is:

```
% vi cts_buffers.tcl
% vi routing_buffers.tcl
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
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Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source cts_buffers.tcl
=====
      CTS BUFFER / INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 12382 components and 78193 component-terminals.
[INFO ODB-0132] Created 2 special nets and 44830 connections.
[INFO ODB-0133] Created 9885 nets and 33257 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/cts/picorv32.def

CTS Buffers   : 2453
CTS Inverters : 81
```



The TCL script to new added buffers/inverters in routing stage is:

```
puts "====="
puts "ROUTING BUFFER / INVERTER REPORT"
puts "====="
read_lef $::env(MERGED_LEF)
read_def /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
set block [ord::get_db_block]
set buf 0
set inv 0
foreach inst [$block getInsts] {
    set cell [string tolower [[$inst getMaster] getName]]
    if {[string match "*buf*" $cell]} {
        incr buf
    }
    if {[string match "*inv*" $cell]} {
        incr inv
    }
}
puts ""
puts "Routing Buffers : $buf"
puts "Routing Inverters : $inv"
```

The output is:

```
% openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> source routing_buffers.tcl
=====
ROUTING BUFFER / INVERTER REPORT
=====
[INFO ODB-0222] Reading LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0223] Created 14 technology layers
[INFO ODB-0224] Created 25 technology vias
[INFO ODB-0225] Created 441 library cells
[INFO ODB-0226] Finished LEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/tmp/merged.nom.lef
[INFO ODB-0127] Reading DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def
[INFO ODB-0128] Design: picorv32
[INFO ODB-0130] Created 411 pins.
[INFO ODB-0131] Created 22018 components and 116887 component-terminals.
[INFO ODB-0132] Created 2 special nets and 83374 connections.
[INFO ODB-0133] Created 9889 nets and 33407 connections.
[INFO ODB-0134] Finished DEF file: /openlane/designs/riscv/runs/FINAL_DESIGN/results/routing/picorv32.def

Routing Buffers : 2452
Routing Inverters : 81
openroad> exit
```



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After running the verification script, this is the output:

```
openroad> source /openlane/designs/riscv/validation_mod.tcl

===== Floorplan & Std Cell Area =====
Core Area   : Width=488.52000000000004 um Height=340.0 um Aspect=0.6959796937689346 Area=166096.800000000002
Die Area    : Width=499.895 um Height=363.96 um Aspect=0.7280728953080147 Area=181941.7842
Std Cell Area : 100371.264000
Saved floorplan_stdcell_report.csv

===== Design Utilization Report =====
Floorplan      | Log file missing
Global_Placement | 62
Detailed_Placement | 62
CTS            | 64
Post_CTS_Resizer | 64
Global_Routing  | 65
Saved design_utilization_report.csv

===== Power Report =====
Synthesis      | 2-sta.log           | 3.28e-02
Global_Placement | 8-gpl_sta.log       | 3.85e-02
Detailed_Placement | 11-dpl_sta.log      | 3.67e-02
CTS            | 13-cts_sta.log       | 4.34e-02
Post_CTS_Resizer | 17-rsz_timing_sta.log | 4.09e-02
Global_Routing  | 20-grt_sta.log       | 3.71e-02
Saved power_report.csv

Detected Clock Period = 6.0 ns

-----
Stage          Clock  InputDelay  Input%  OutputDelay  Output%
-----
Synthesis      6.00   3.0         50.00   3.0          50.00
Global_Placement 6.00   3.0         50.00   3.0          50.00
Detailed_Placement 6.00   3.0         50.00   3.0          50.00
CTS            6.00   3.0         50.00   3.0          50.00
-----

-----
Stage          Clock  InputDelay  Input%  OutputDelay  Output%
-----
Synthesis      6.00   3.0         50.00   3.0          50.00
Global_Placement 6.00   3.0         50.00   3.0          50.00
Detailed_Placement 6.00   3.0         50.00   3.0          50.00
CTS            6.00   3.0         50.00   3.0          50.00
Post_CTS_Resizer 6.00   3.0         50.00   3.0          50.00
Global_Routing  6.00   3.0         50.00   3.0          50.00
Saved in_out_delay_report.csv

===== Worst Setup / Hold Slack Report =====
Synthesis      | -1.28 | -0.48
Global_Placement | -1.40 | -0.48
Detailed_Placement | -1.60 | -0.46
CTS            | -1.61 | -0.46
Post_CTS_Resizer | -1.54 | -0.45
Global_Routing  | -1.50 | -0.47
Saved worst_slack_report.csv

===== Clock Uncertainty Report =====
Synthesis      | 0.90 | 0.90
Global_Placement | 0.90 | 0.90
Detailed_Placement | 0.90 | 0.90
CTS            | 0.90 | 1.59
Post_CTS_Resizer | 0.90 | 1.39
Global_Routing  | 0.90 | 1.44
Saved uncertainty_report.csv

===== Register-to-Register Worst Slack Report =====
Synthesis      | Setup | _15922_ | _16079_ | -1.11
Synthesis      | Hold  | _15920_ | _15920_ | -0.48
Global_Placement | Setup | _15922_ | _16079_ | -1.38
Global_Placement | Hold  | _15920_ | _15920_ | -0.48
Detailed_Placement | Setup | _16114_ | _15918_ | -0.97
Detailed_Placement | Hold  | _15920_ | _15920_ | -0.46
CTS            | Setup | _16114_ | _15918_ | -0.97
CTS            | Hold  | _15920_ | _15920_ | -0.46
Post_CTS_Resizer | Setup | _15921_ | _16079_ | -0.79
Post_CTS_Resizer | Hold  | _15957_ | _15957_ | -0.45
Global_Routing  | Setup | _15921_ | _16079_ | -0.30
Global_Routing  | Hold  | _15920_ | _15920_ | -0.47

=====
Saved reg2reg_worst_slack.csv
-----
```



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```
===== Validation Script Completed =====  
Generated CSV Files:  
  floorplan_stdcell_report.csv  
  design_utilization_report.csv  
  power_report.csv  
  in_out_delay_report.csv  
  worst_slack_report.csv  
  uncertainty_report.csv  
  reg2reg_worst_slack.csv  
=====  
openroad> █
```